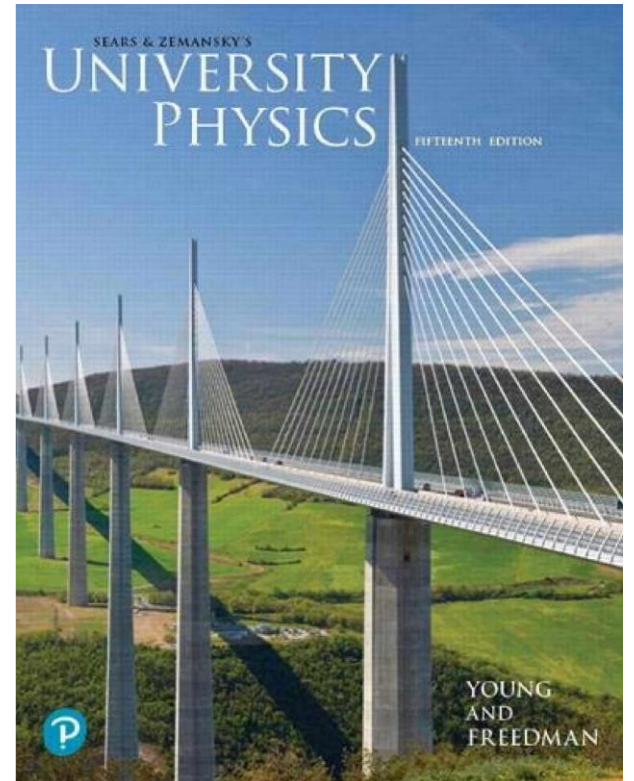


普通物理I PHYS1181

振动 Oscillation



本节课（振动）：

A. 振动的基本概念

B. 简谐振动

C. 简谐振动的能量

D. 扭摆

E. 单摆

F. 复摆

CHAPTER 14 SUMMARY

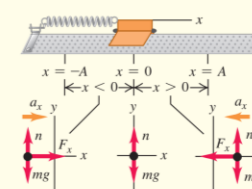
Periodic motion: Periodic motion is motion that repeats itself in a definite cycle. It occurs whenever an object has a stable equilibrium position and a restoring force that acts when the object is displaced from equilibrium. Period T is the time for one cycle. Frequency f is the number of cycles per unit time. Angular frequency ω is 2π times the frequency. (See Example 14.1.)

$$f = \frac{1}{T} \quad T = \frac{1}{f}$$

(14.1)

$$\omega = 2\pi f = \frac{2\pi}{T}$$

(14.2)



Simple harmonic motion: If the restoring force F_x in periodic motion is directly proportional to the displacement x , the motion is called simple harmonic motion (SHM). In many cases this condition is satisfied if the displacement from equilibrium is small. The angular frequency, frequency, and period in SHM do not depend on the amplitude but on only the mass m and force constant k . The displacement, velocity, and acceleration in SHM are sinusoidal functions of time; the amplitude A and phase angle ϕ of the oscillation are determined by the initial displacement and velocity of the object. (See Examples 14.2, 14.3, 14.6, and 14.7.)

$$F_x = -kx$$

(14.3)

$$a_x = \frac{F_x}{m} = -\frac{k}{m}x$$

(14.4)

$$\omega = \sqrt{\frac{k}{m}}$$

(14.10)

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

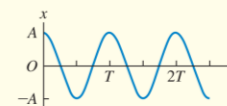
(14.11)

$$T = \frac{1}{f} = 2\pi \sqrt{\frac{m}{k}}$$

(14.12)

$$x = A \cos(\omega t + \phi)$$

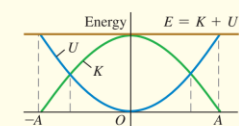
(14.13)



Energy in simple harmonic motion: Energy is conserved in SHM. The total energy can be expressed in terms of the force constant k and amplitude A . (See Examples 14.4 and 14.5.)

$$E = \frac{1}{2}mv_c^2 + \frac{1}{2}kx^2 \\ = \frac{1}{2}kA^2 = \text{constant}$$

(14.21)



Angular simple harmonic motion: In angular SHM, the frequency and angular frequency are related to the moment of inertia I and the torsion constant κ .

$$\omega = \sqrt{\frac{\kappa}{I}} \quad \text{and}$$

(14.24)

$$f = \frac{1}{2\pi} \sqrt{\frac{\kappa}{I}}$$



Simple pendulum: A simple pendulum consists of a point mass m at the end of a massless string of length L . Its motion is approximately simple harmonic for sufficiently small amplitude; the angular frequency, frequency, and period then depend on only g and L , not on the mass or amplitude. (See Example 14.8.)

$$\omega = \sqrt{\frac{g}{L}}$$

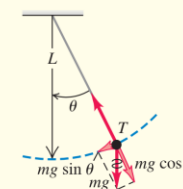
(14.32)

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$$

(14.33)

$$T = \frac{2\pi}{\omega} = \frac{1}{f} = 2\pi \sqrt{\frac{L}{g}}$$

(14.34)



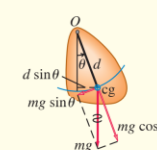
Physical pendulum: A physical pendulum is any object suspended from an axis of rotation. The angular frequency and period for small-amplitude oscillations are independent of amplitude but depend on the mass m , distance d from the axis of rotation to the center of gravity, and moment of inertia I about the axis. (See Examples 14.9 and 14.10.)

$$\omega = \sqrt{\frac{mgd}{I}}$$

(14.38)

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

(14.39)



本节课（振动）：

G. 阻尼振动

Q值

H. 受迫振动、共振

I. 简谐振动的合成

同方向同频率

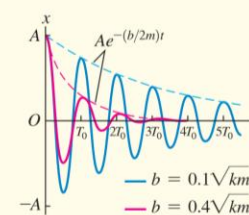
同方向不同频率、拍

垂直方向、李萨如图形

Damped oscillations: When a force $F_x = -bv_x$ is added to a simple harmonic oscillator, the motion is called a damped oscillation. If $b < 2\sqrt{km}$ (called underdamping), the system oscillates with a decaying amplitude and an angular frequency ω' that is lower than it would be without damping. If $b = 2\sqrt{km}$ (called critical damping) or $b > 2\sqrt{km}$ (called overdamping), when the system is displaced it returns to equilibrium without oscillating.

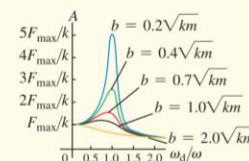
$$x = Ae^{-(b/2m)t} \cos(\omega't + \phi) \quad (14.42)$$

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \quad (14.43)$$



Forced oscillations and resonance: When a sinusoidally varying driving force is added to a damped harmonic oscillator, the resulting motion is called a forced oscillation or driven oscillation. The amplitude is a function of the driving frequency ω_d and reaches a peak at a driving frequency close to the natural frequency of the system. This behavior is called resonance.

$$A = \frac{F_{\max}}{\sqrt{(k - m\omega_d^2)^2 + b^2\omega_d^2}} \quad (14.46)$$



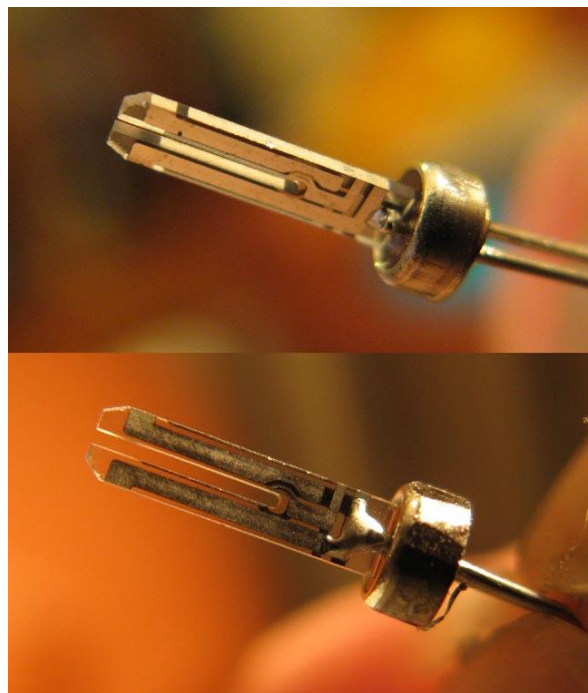
振动是自然界极为普遍的现象。



石英手表中的晶振

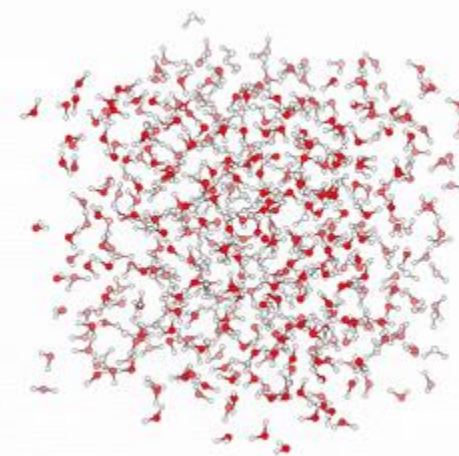
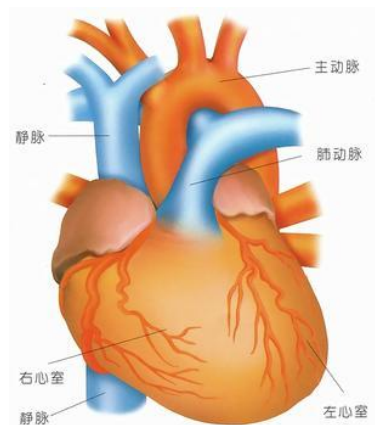
实际上就是一个石英做的小音叉 (tuning fork)

只要不是绝对零度，物质中所有的分子、所有的原子就都在不断的振动



人体中的振动

心脏的搏动
声带、耳膜的振动



水团簇的分子动力学模拟

A. 振动（周期运动）

什么是振动？

周期性的运动 periodic motion

振：vibration（振动），oscillation（振荡）

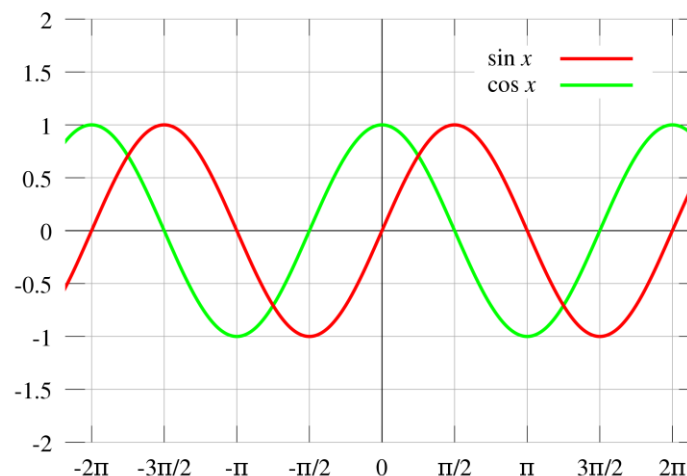
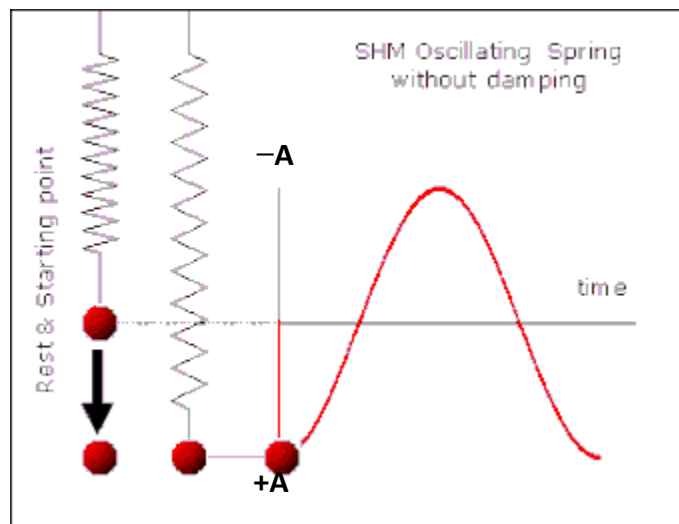
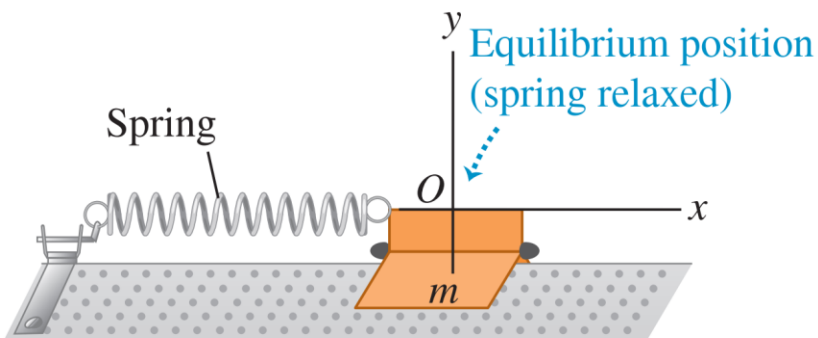
振动 $\neq$ 震动

振荡 $\neq$ 震荡

地震
手机震动
古龙小说：虎躯一震

股市震荡

震：quake, shake



B. 简谐振动 (Simple Harmonic oscillation)

B1: 简谐运动 (simple harmonic motion) : 弹性力 F 与偏离平衡位置的位移 x 成正比

理想弹簧的弹性力，满足线性的胡克定律：

Restoring force exerted by an ideal spring

$$F_x = -kx$$

x -component of force
Displacement
Force constant of spring

质量为 m 物体做简谐振动的加速度方程：

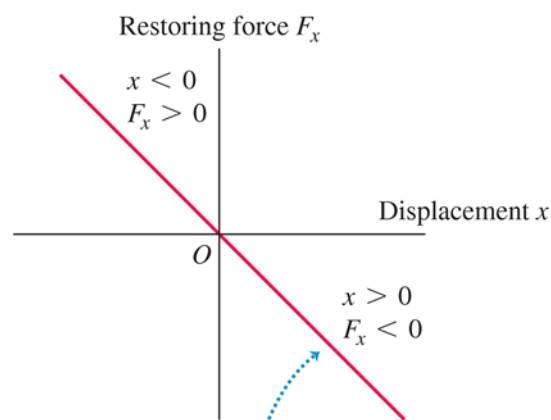
Equation for simple harmonic motion

$$a_x = \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

x -component of acceleration
Force constant of restoring force
Displacement
Mass of object
Second derivative of displacement

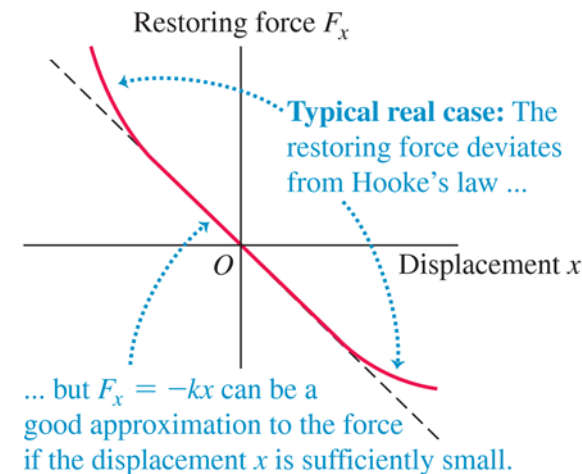
- 简谐振动中，周期与频率不依赖于幅度 A .

理想弹簧



The restoring force exerted by an idealized spring is directly proportional to the displacement (Hooke's law, $F_x = -kx$): the graph of F_x versus x is a straight line.

实际弹簧



B2: 简谐振动的描述

由简谐运动的加速度方程: $F_x = ma = m \frac{d^2x}{dt^2} = -kx \quad \Rightarrow \quad \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$

定义**角频率** ω :

Angular frequency for simple harmonic motion $\omega = \sqrt{\frac{k}{m}}$ Force constant of restoring force Mass of object

$$\omega^2 = \frac{k}{m}$$

可得: $\frac{d^2x}{dt^2} + \omega^2x = 0$ (简谐振动的特征方程)

解此微分方程, 可得**简谐运动表达式**

Displacement in simple harmonic motion as a function of time

Amplitude

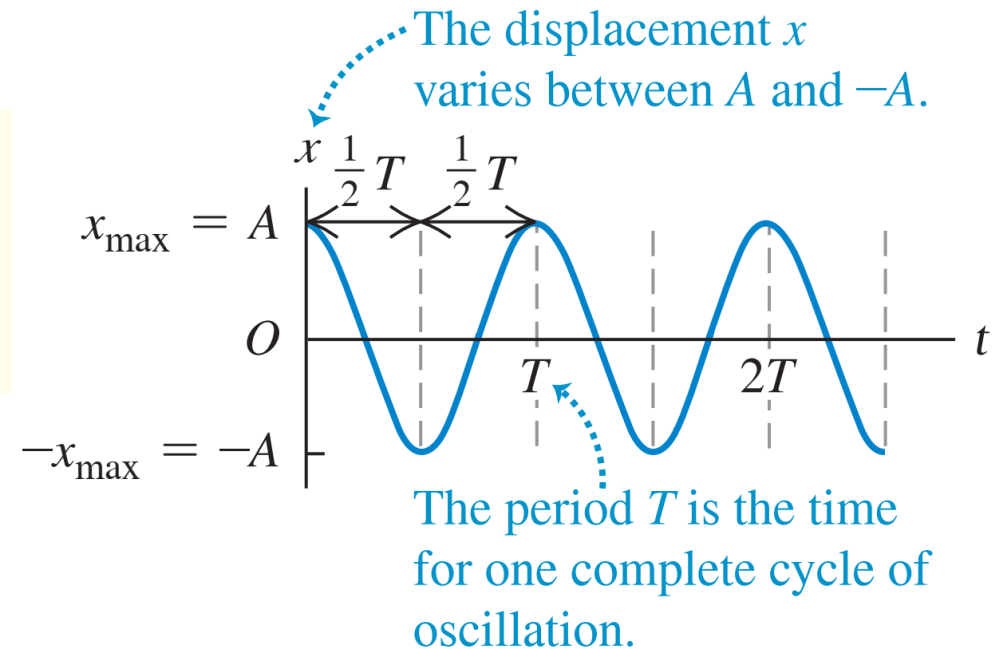
Time

Phase angle

$$x = A \cos(\omega t + \phi)$$

Angular frequency $= \sqrt{k/m}$

$$\omega = \frac{2\pi}{T} = 2\pi f \quad T = 2\pi \sqrt{\frac{m}{k}}$$



[例题] 证明图示系统的振动为简谐振动。其频率为

证：

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k_1 k_2}{(k_1 + k_2)m}}$$

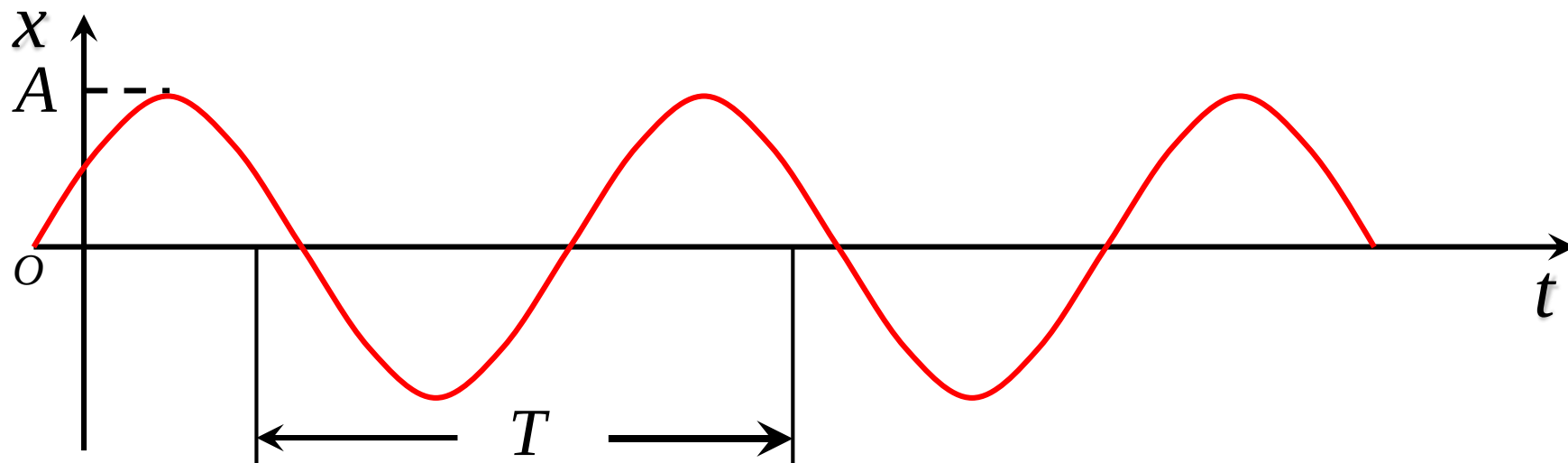
B3: 描述简谐振动的物理量

$$x = A \cos(\omega t + \varphi_0)$$

1. 振幅(amplitude) : A (即最大位移, $x = \pm A$)
2. 周期(period) T : 完成一次全振动所经历的时间。 T 的单位: 秒(s)

频率 f : 单位时间内完成全振动的次数。 $f = 1/T$ 。 f 的单位: 赫兹(Hz)

ω : 角频率 (或称圆频率) ω 的单位: 弧度每秒(rad/s)



3. 相位 (phase) : $(\omega t + \varphi_0)$ ——描述振动状态

初相位 (initial phase) : φ_0

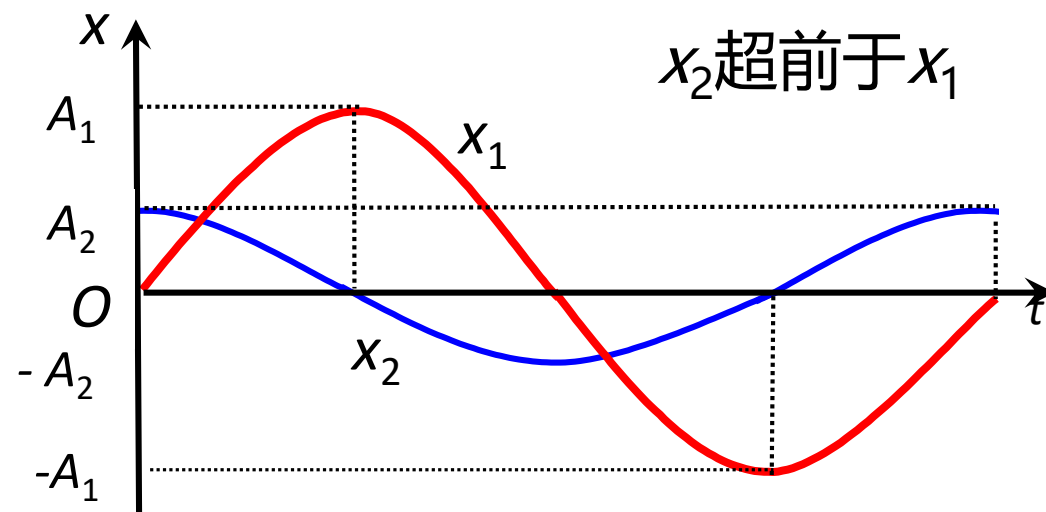
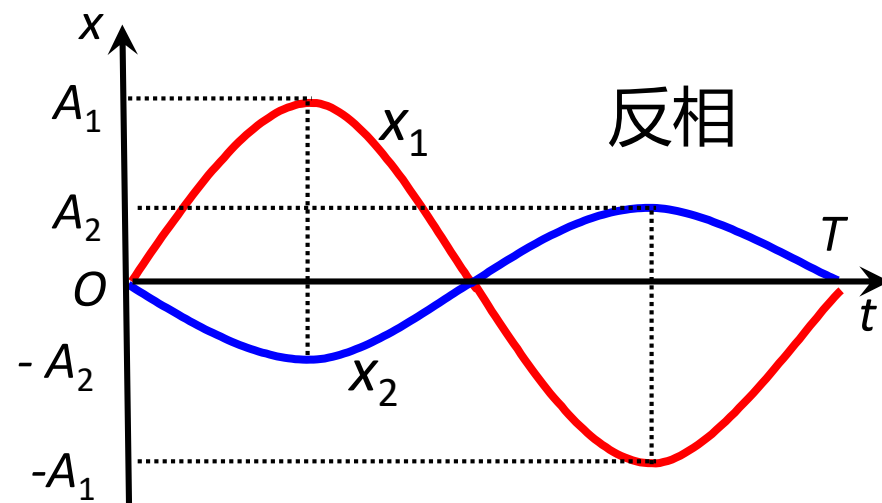
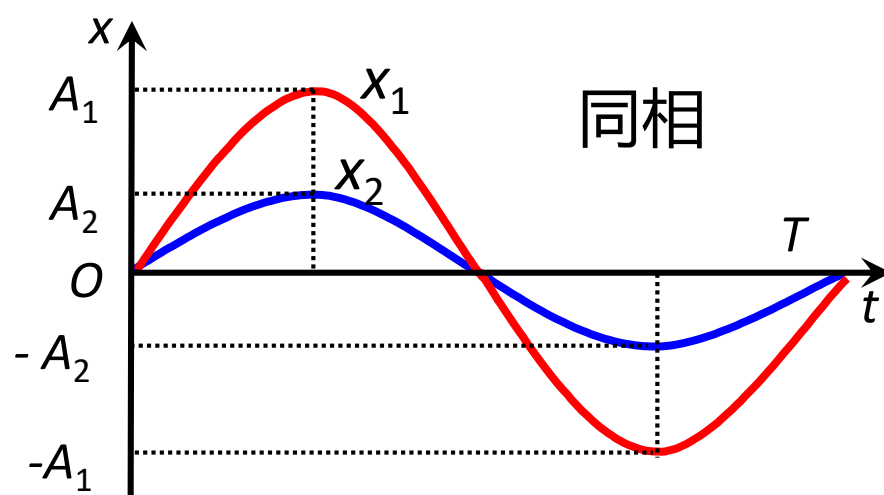
相位差: $\Delta\varphi = (\omega_2 t + \varphi_{20}) - (\omega_1 t + \varphi_{10})$

对两同频率的谐振动 $\Delta\varphi = \varphi_{20} - \varphi_{10}$ 初相差

当 $\Delta\varphi = \pm 2k\pi$, ($k=0,1,2,\dots$), 两振动步调相同, 称同相。

当 $\Delta\varphi = \pm (2k+1)\pi$, ($k=0,1,2,\dots$), 两振动步调相反, 称反相。

若 $0 < \varphi_{20} - \varphi_{10} < \pi$, 则 x_2 比 x_1 较早达到正最大, 称 x_2 比 x_1 超前 (或 x_1 比 x_2 落后)。



简谐运动的速度方程、加速度方程：

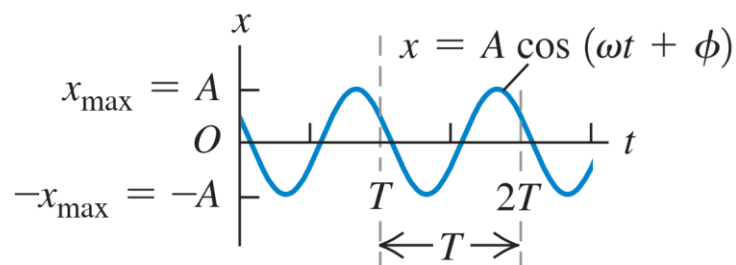
振动速度： $v = \frac{dx}{dt} = -\omega A \sin(\omega t + \varphi_0) = v_m \cos(\omega t + \varphi_0 + \frac{\pi}{2})$

其中： $v_m = \omega A$ 称为 “速度幅值”

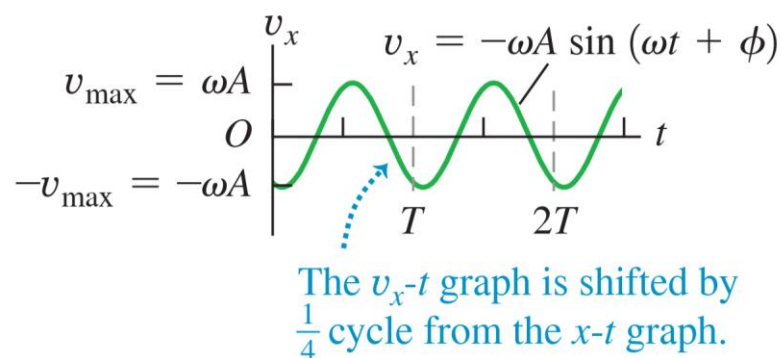
振动加速度： $a = \frac{dv}{dt} = -\omega^2 A \cos(\omega t + \varphi_0) = a_m \cos(\omega t + \varphi_0 \pm \pi)$

其中： $a_m = \omega^2 A$ 称为 “加速度幅值”

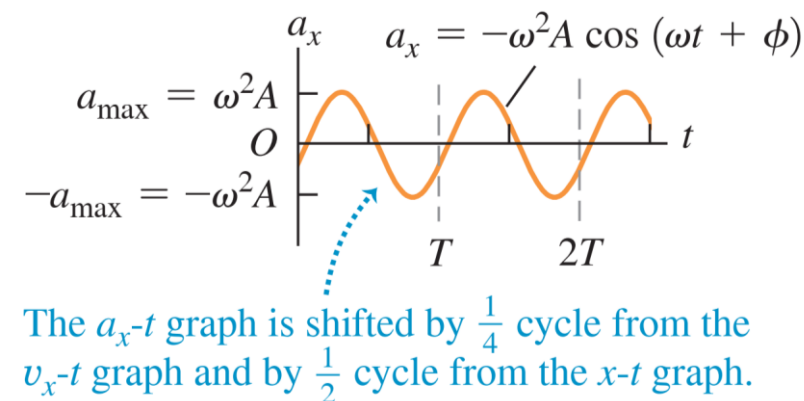
(a) Displacement x as a function of time t



(b) Velocity v_x as a function of time t



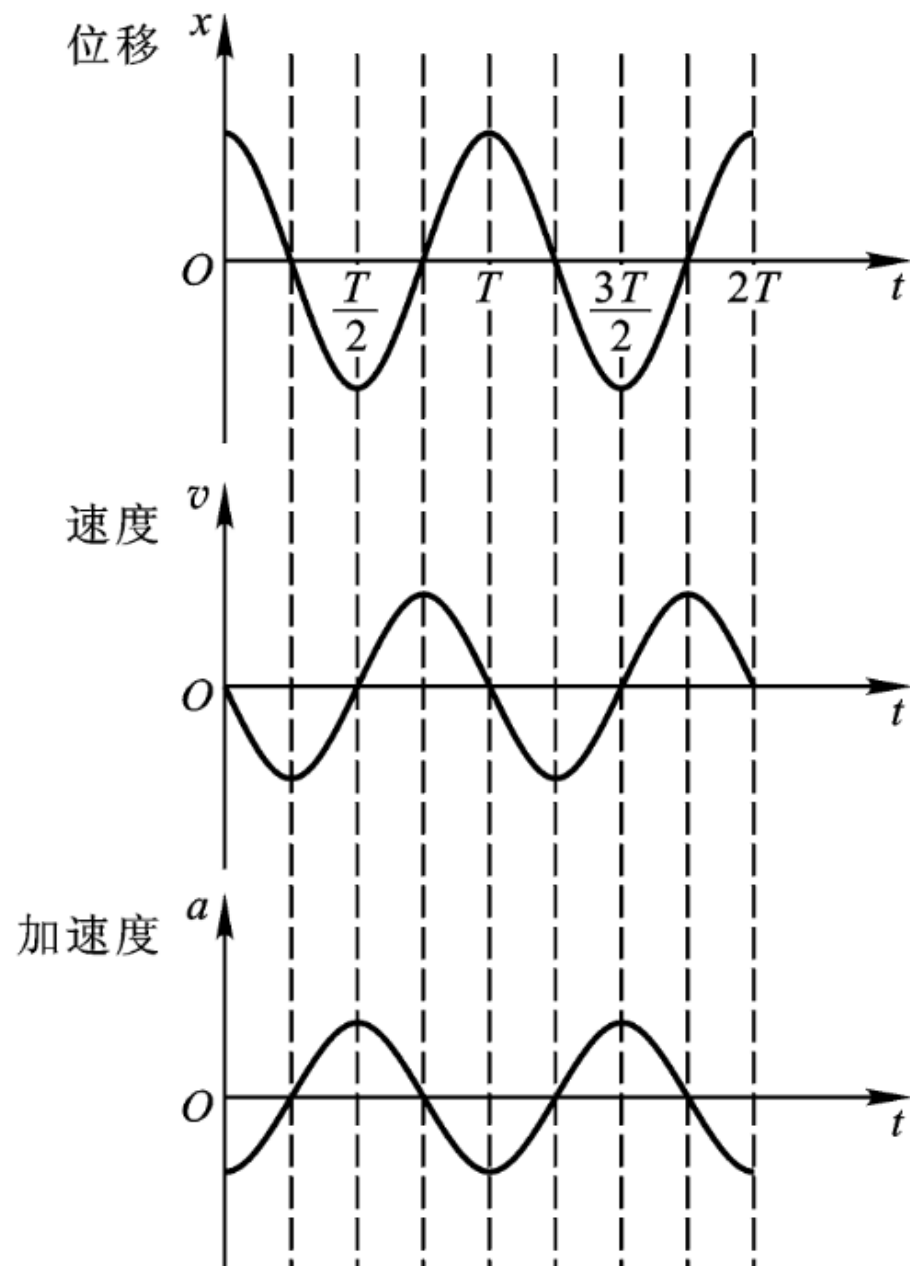
(c) Acceleration a_x as a function of time t



$$v = v_m \cos(\omega t + \varphi_0 + \frac{\pi}{2})$$

$$a = a_m \cos(\omega t + \varphi_0 \pm \pi)$$

速度相位比位移相位超前 $\pi/2$ 。
加速度与位移反相位。

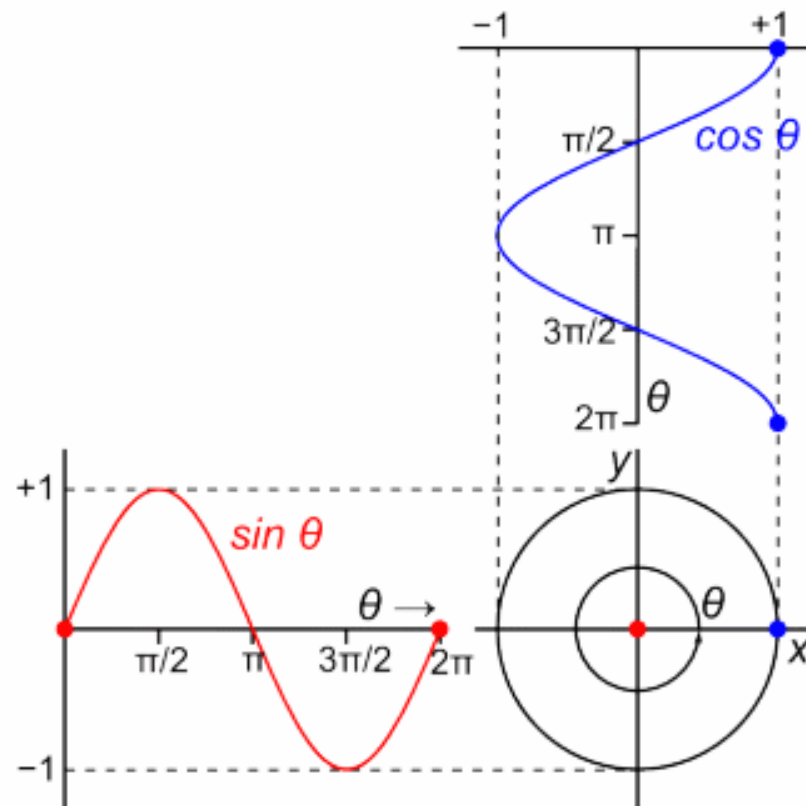
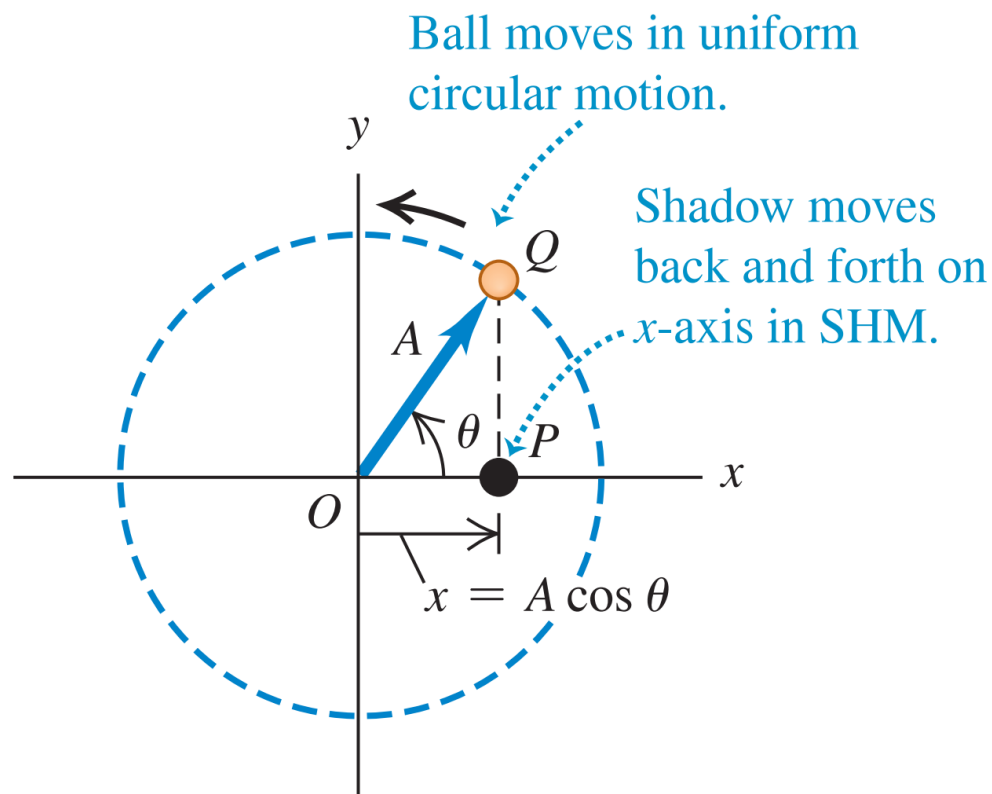


简谐运动方程也可以利用其和匀速圆周运动的关系来描述：

一个匀速圆周运动的物体在其直径上的投影 x 随时间的变化，即为简谐运动。

因此有： $x = A \cos \theta$ 其中， θ 为任一时刻 t 物体的角位置 $\theta = \omega t + \varphi$ ， φ 为起始角位置。

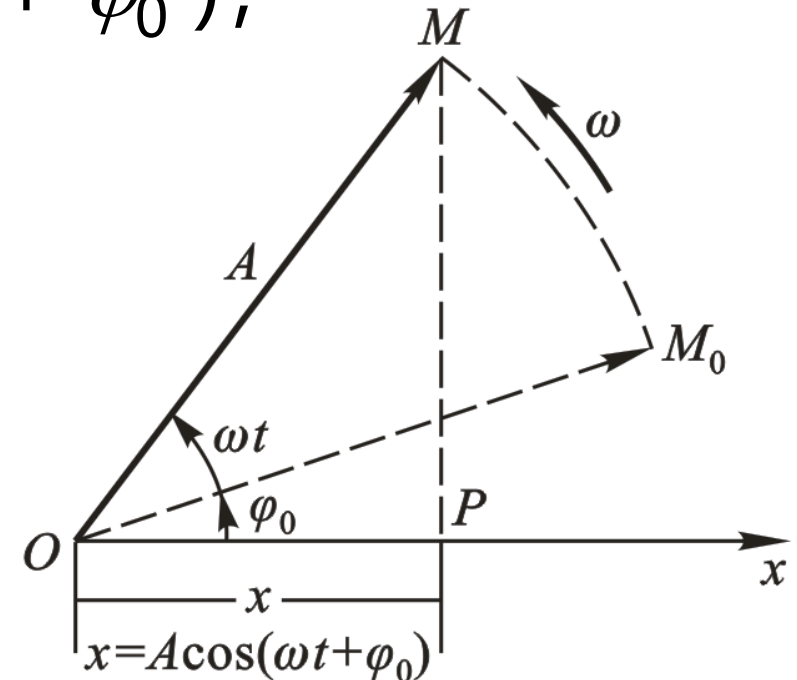
则立刻可得： $x = A \cos (\omega t + \varphi)$



B4: 谐振动的旋转矢量图示法

旋转矢量 $\vec{A}$ 绕原点逆时针方向旋转

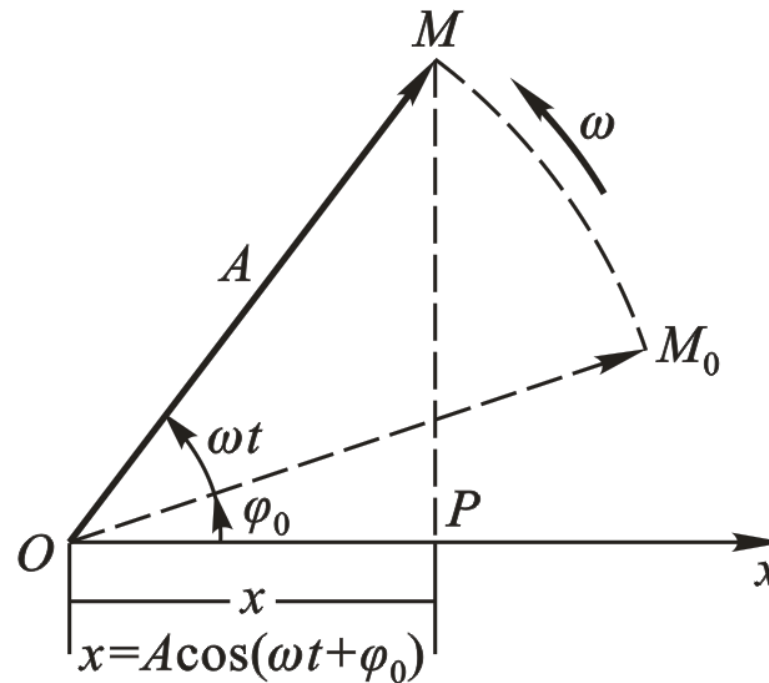
- 旋转矢量 $\vec{A}$ 的模即为简谐振动的**振幅**。
- 旋转矢量 $\vec{A}$ 的角速度 ω 即为振动的**角频率**。
- 旋转矢量 $\vec{A}$ 与 x 轴的夹角 $(\omega t + \varphi_0)$, 为简谐振动的**相位**。
- $t=0$ 时, $\vec{A}$ 与 x 轴的夹角 φ_0 即为简谐振动的**初相位**。

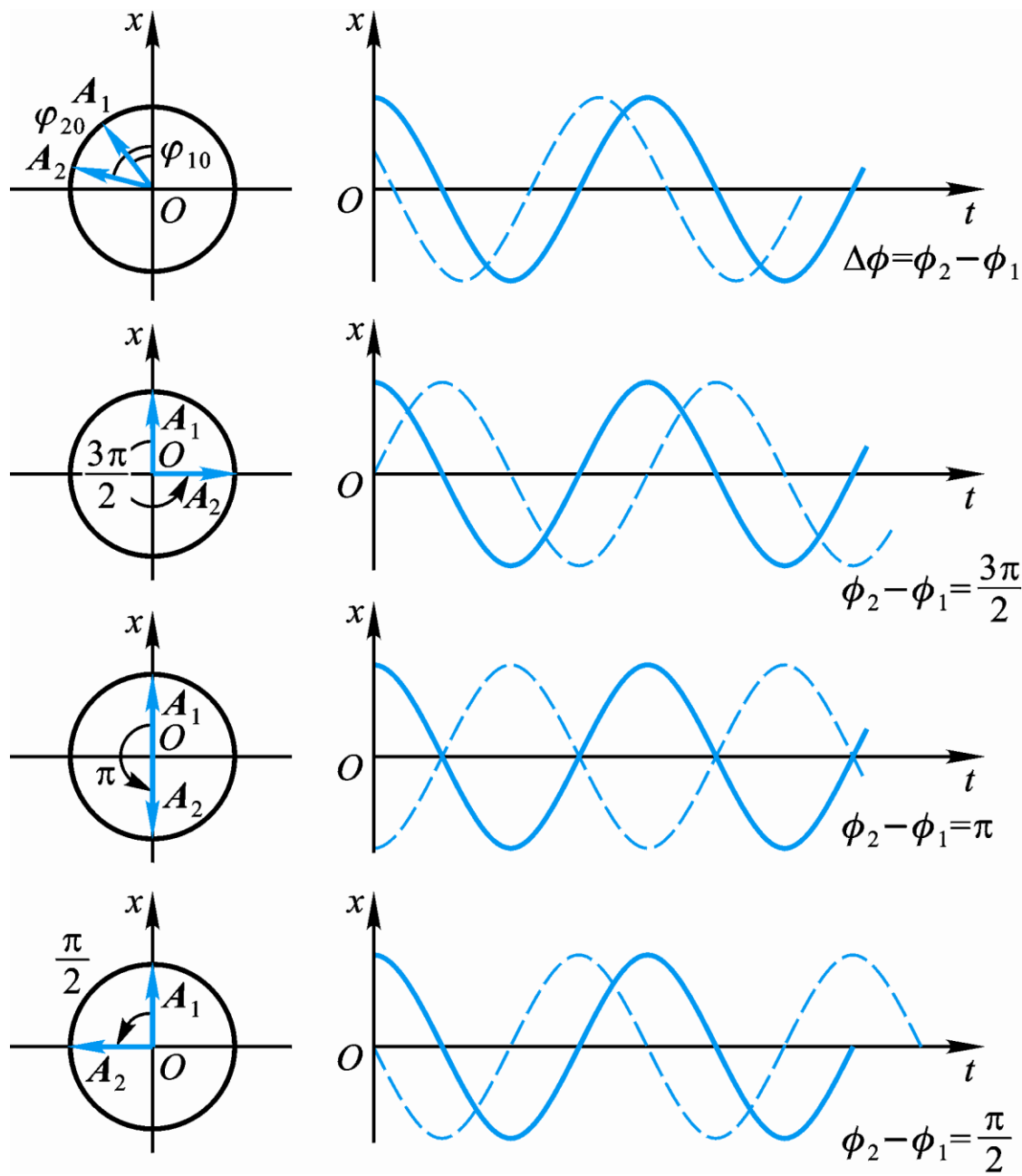


旋转矢量 $\vec{A}$ 的端点在 x 轴上的投影点 P 的位移：

$$x = A \cos(\omega t + \varphi_0)$$

➤ 投影点 P 的运动为简谐振动。





两简谐振动同频率同振幅初相位不同

[例题] 一质点沿x轴作简谐振动，振幅为12cm，周期为2s。当 $t=0$ 时，位移为6cm，且向x轴正方向运动。求：1. 振动式。2. $t=0.5\text{s}$ 时，质点的位置、速度和加速度。3. 如果在某时刻质点位于 $x=-6\text{cm}$ ，且向x轴负方向运动，求从该位置回到平衡位置所需要的时间。

C. 简谐振动的能量

以水平弹簧振子为例 $x = A \cos(\omega t + \varphi_0)$

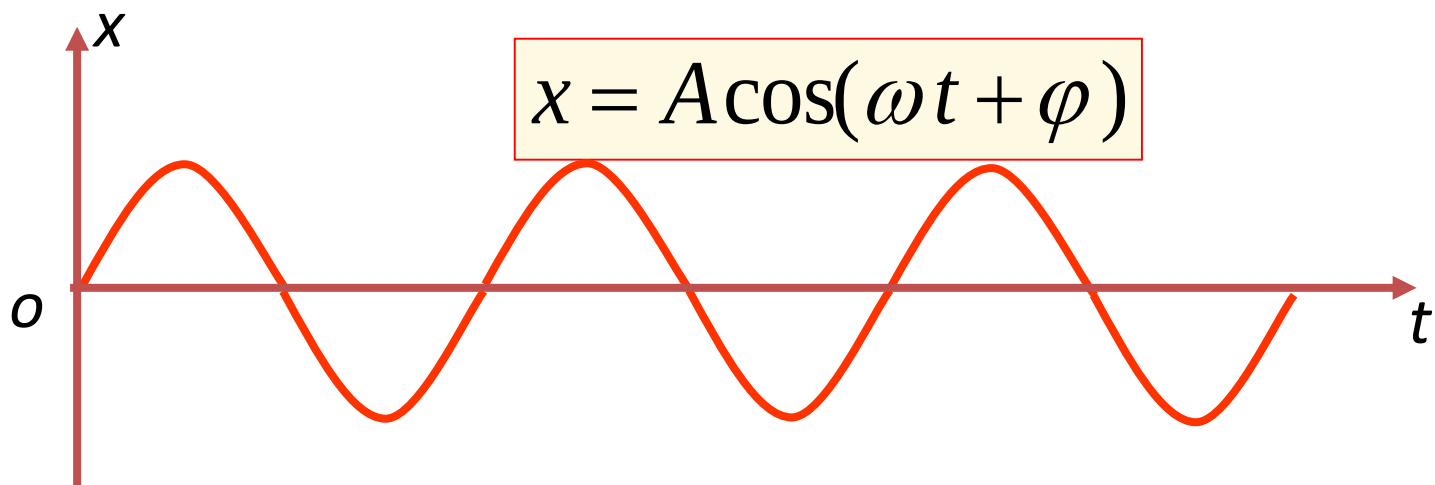
$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \varphi_0)$$

动能: $E_k = \frac{1}{2} m v^2 = \frac{1}{2} k A^2 \sin^2(\omega t + \varphi_0)$

势能: $E_p = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \varphi_0)$

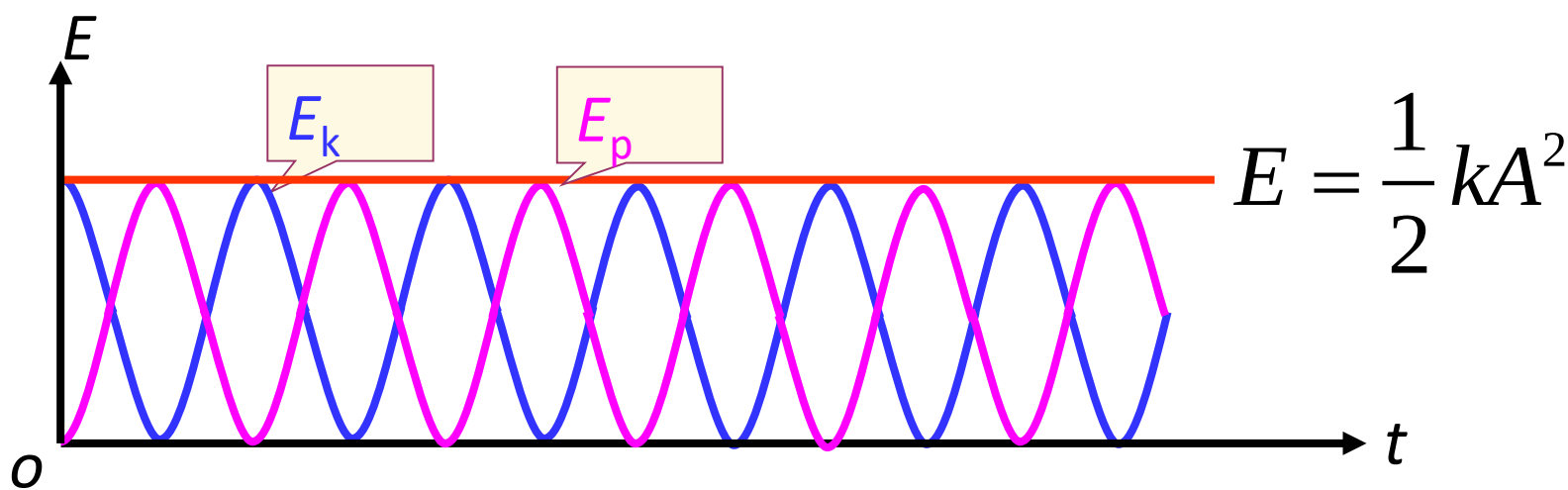
所以总的机械能: $E = E_k + E_p = \frac{1}{2} k A^2$

➤ 简谐振动系统机械能守恒。



➤ 振子在振动过程中，动能和势能分别随时间变化，但任一时刻**总机械能**保持不变。

$$v_x = \pm \sqrt{\frac{k}{m}} \sqrt{A^2 - x^2}$$



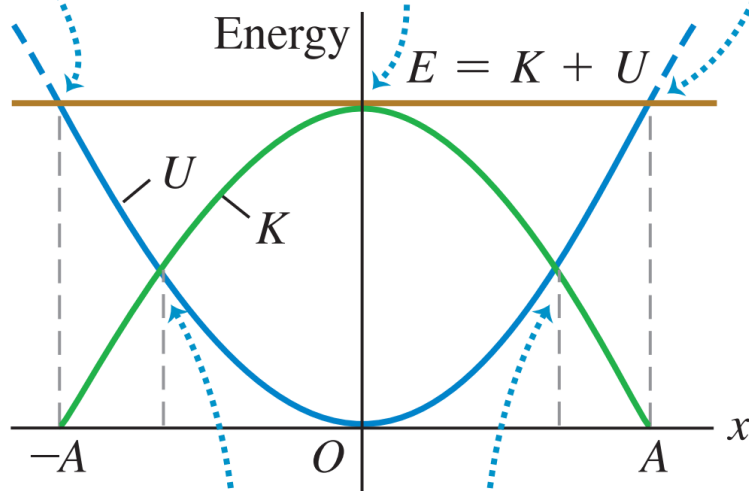
➤ 动能和势能的变化频率是弹簧振子振动频率的**两倍**。

➤ 谐振动的总能量与**振幅的平方**成正比。（适合于任何谐振系统）

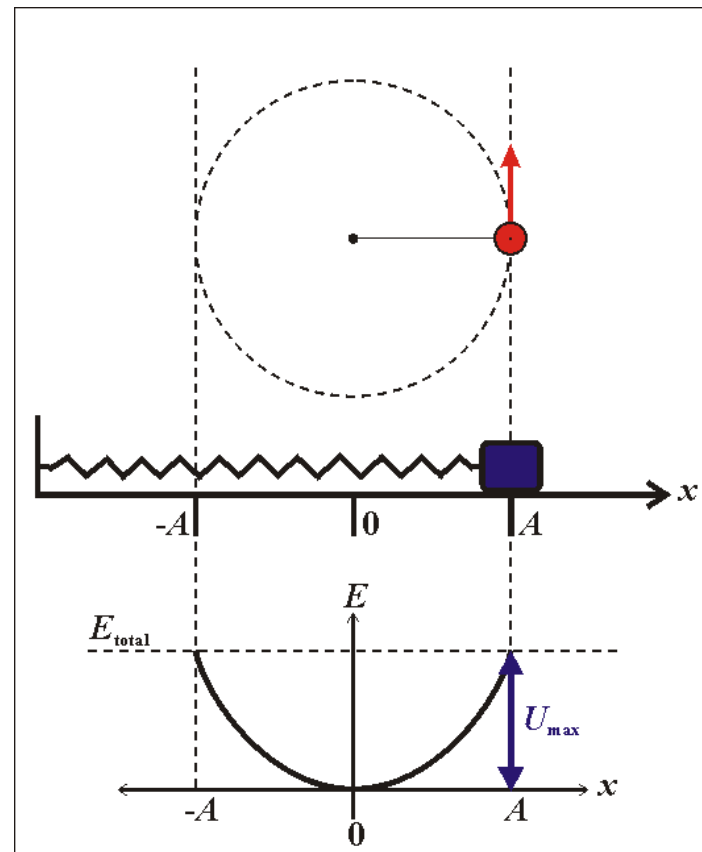
简谐振动的总机械能守恒 $E = \frac{1}{2}kA^2$, 只和最大振幅A有关

At $x = \pm A$ the energy is all potential; $K = 0$.

At $x = 0$ the energy is all kinetic; $U = 0$.

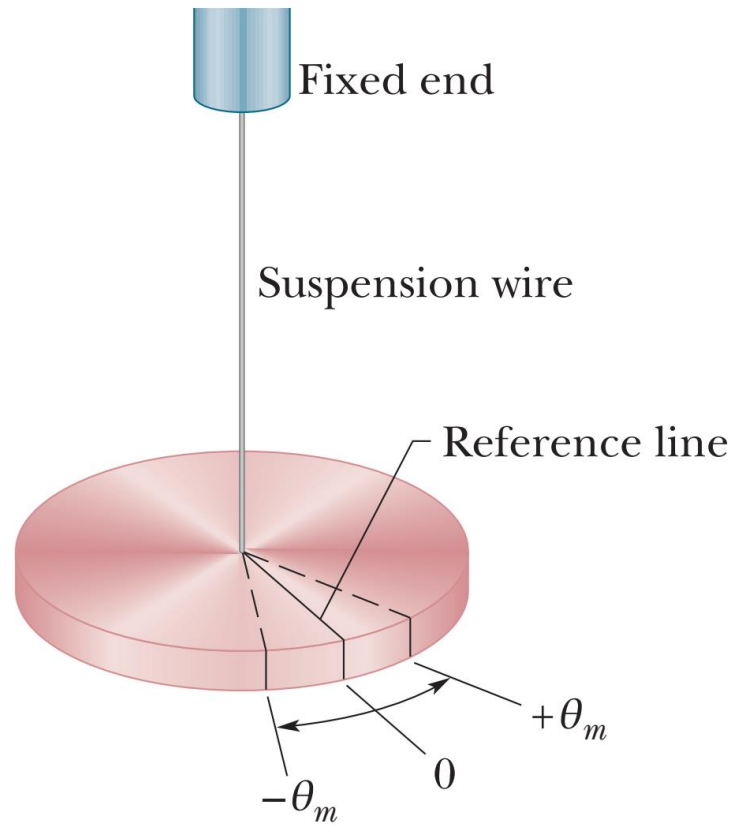


At these points the energy is half kinetic and half potential.



D. 扭摆 (torsion pendulum)

扭摆是一种角简谐振动 (angular simple harmonic motion)



扭矩: $\tau = -\kappa\theta$.

$$\theta = \Theta \cos(\omega t + \phi)$$

κ 为扭转常数 Θ 为最大角位移 (角幅度)

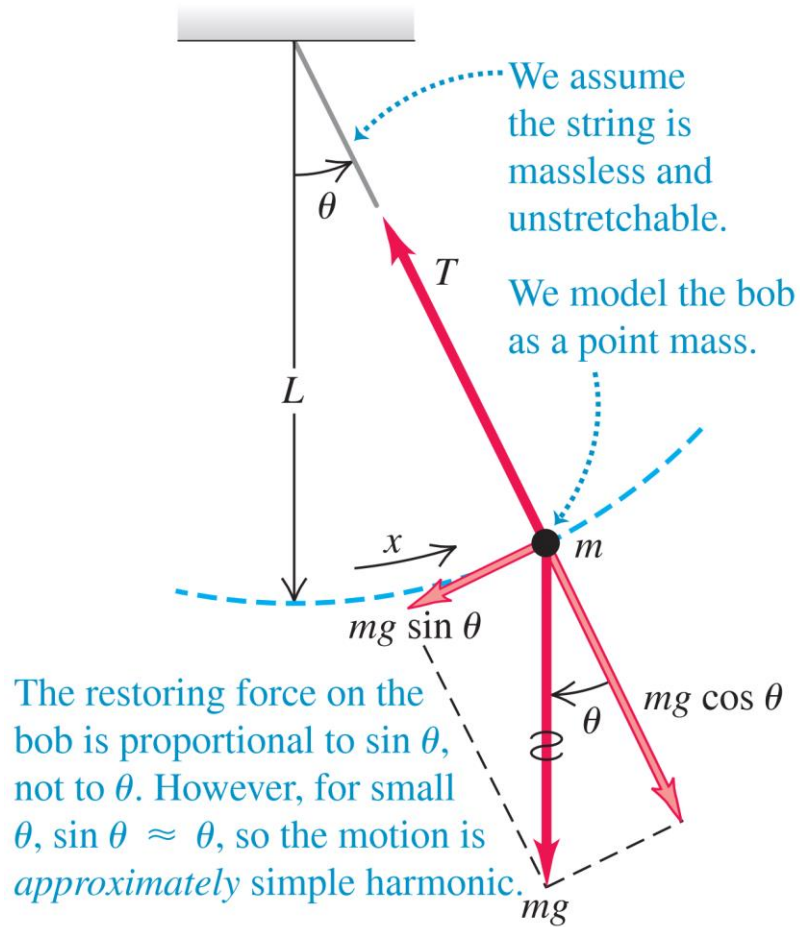
Angular simple harmonic motion

Angular frequency $\omega = \sqrt{\frac{\kappa}{I}}$ and Frequency $f = \frac{1}{2\pi} \sqrt{\frac{\kappa}{I}}$

Torsion constant divided by moment of inertia

I 为圆盘的转动惯量

E. 单摆 (simple pendulum)



$$F_{\theta} = -mg \sin \theta$$

振幅很小时, $\sin \theta \sim \theta$, 近似为简谐运动:

$$F_{\theta} = -mg\theta = -mg \frac{x}{L} = -\frac{mg}{L}x$$

角频率为:

Angular frequency of simple pendulum, small amplitude

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{mg/L}{m}} = \sqrt{\frac{g}{L}}$$

Acceleration due to gravity

Pendulum mass (cancels)

Pendulum length

周期为:

Period of simple pendulum, small amplitude

$$T = \frac{2\pi}{\omega} = \frac{1}{f} = 2\pi \sqrt{\frac{L}{g}}$$

Angular frequency

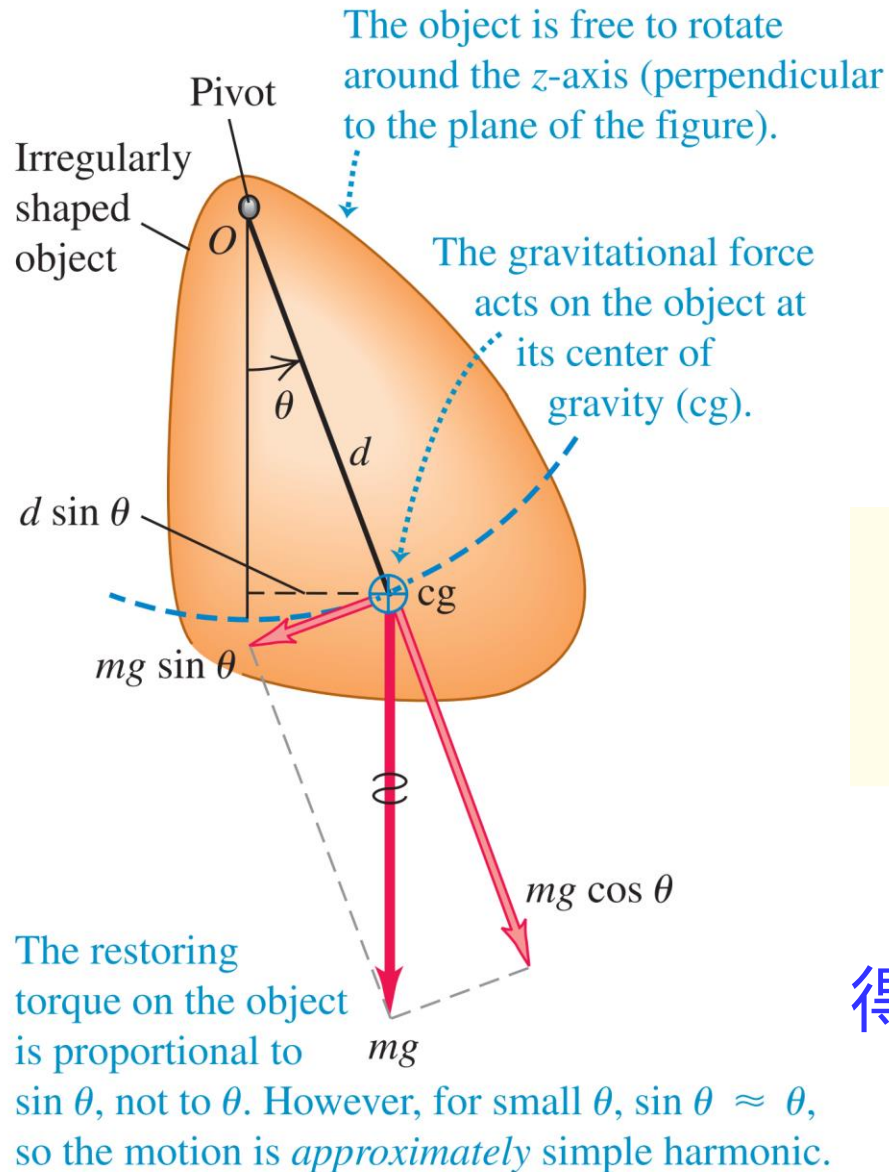
Frequency

Pendulum length

Acceleration due to gravity

F. 复摆 (physical pendulum)

$$\frac{d^2\theta}{dt^2} + \omega^2 \sin \theta = 0$$



$$\tau_z = -(mg)(d \sin \theta)$$

振幅很小时, $\sin \theta \sim \theta$, 近似为简谐运动:

$$\tau_z = -(mgd)\theta$$

角频率为:

Angular frequency of physical pendulum, small amplitude

$$\omega = \sqrt{\frac{mgd}{I}}$$

Mass

Acceleration due to gravity

Distance from rotation axis to center of gravity

Moment of inertia

周期为:

得:

Period of physical pendulum, small amplitude

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

Moment of inertia

Distance from rotation axis to center of gravity

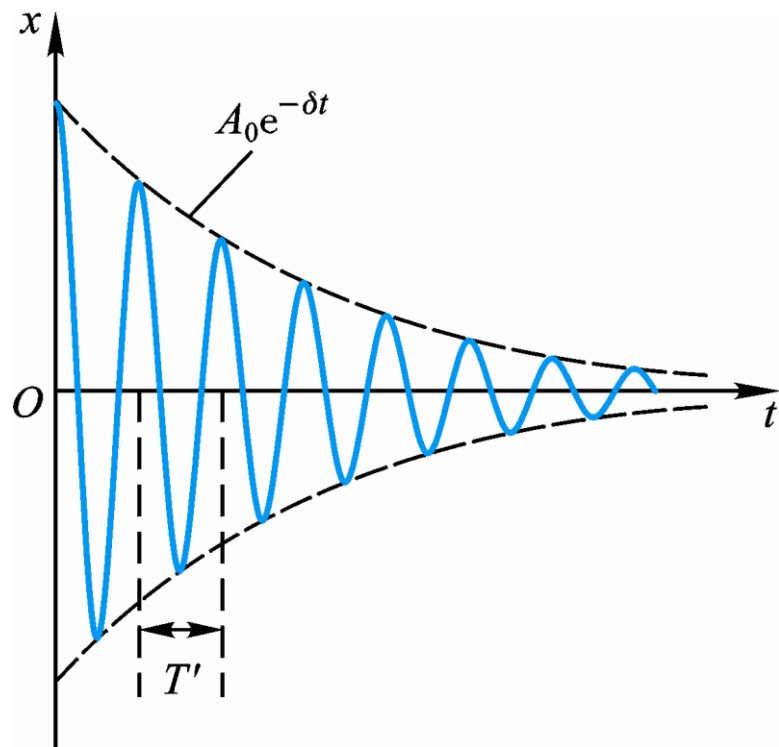
Mass

Acceleration due to gravity

G. 阻尼振动 (damped oscillation)

振动物体不受任何阻力的影响，只在回复力作用下所做的振动，称为**无阻尼自由振动**。

振动物体在回复力和阻力作用下的振动称为**阻尼振动**。



阻尼：消耗振动系统能量的原因。

阻尼种类：摩擦阻尼 辐射阻尼

对在流体(液体、气体)中运动的物体，当物体速度较小时，阻力大小正比于速度，且方向相反，表示为

$$F_f = -\gamma v = -\gamma \frac{dx}{dt} \quad \gamma : \text{阻力系数}$$

阻尼振动方程： $m \frac{d^2 x}{dt^2} = -kx - \gamma \frac{dx}{dt}$

引入 阻尼因子 $\delta = \frac{\gamma}{2m}$ 固有频率 $\omega_0 = \sqrt{\frac{k}{m}}$

$$\Rightarrow \frac{d^2 x}{dt^2} + 2\delta \frac{dx}{dt} + \omega_0^2 x = 0$$

在小阻尼条件下 ($\delta < \omega_0$) , 微分方程的解为

$$x = A_0 e^{-\delta t} \cos(\omega' t + \varphi'_0)$$

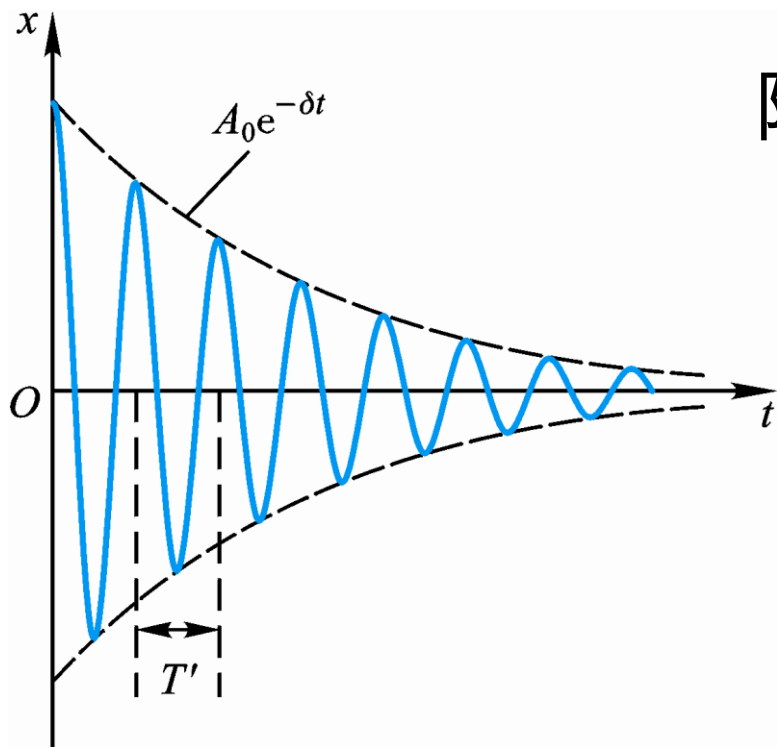
其中 $\omega' = \sqrt{\omega_0^2 - \delta^2}$

A_0 和 φ'_0 为积分常数, 由初始条件决定。

$$x = A_0 e^{-\delta t} \cos(\omega' t + \varphi'_0)$$

余弦项表征了在弹性力和阻力作用下的周期运动；

$A_0 e^{-\delta t}$ 反映了阻尼对振幅的影响。 ——减幅振动



阻尼振动是准周期性运动

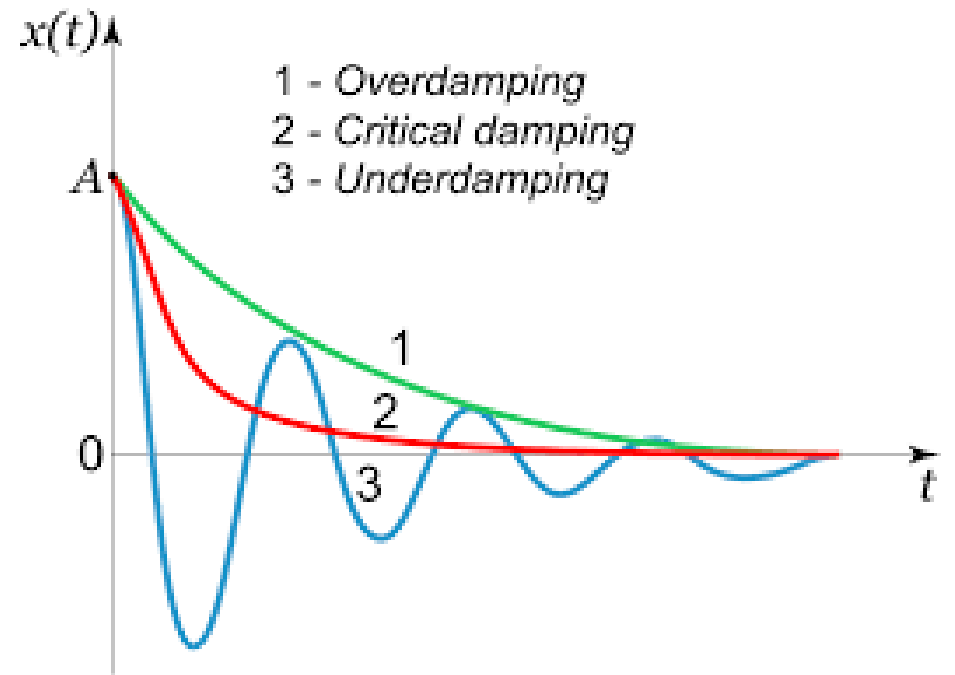
阻尼振动的周期：

$$T' = \frac{2\pi}{\omega'} = \frac{2\pi}{\sqrt{\omega_0^2 - \beta^2}} > \frac{2\pi}{\omega_0}$$

阻尼振动的三种情形：

- 过阻尼 $\delta > \omega_0$
- 阻尼 $\delta < \omega_0$
- 临界阻尼 $\delta = \omega_0$

➤ 临界阻尼物体从静止开始回到平衡位置所需的时间最短



阻尼项的数学体现* (速度)如何衰减?

例：考虑以 v_0 运动的物体在只有粘滞阻力作用时的运动情况

$$m \frac{d^2 x}{dt^2} = -b \frac{dx}{dt} \quad \text{即} \quad \frac{d^2 x}{dt^2} + 2\delta \frac{dx}{dt} = 0$$
$$\therefore \frac{dv}{dt} = -2\delta v \quad \frac{dv}{v} = -2\delta dt$$

两边同时积分得： $\ln v - \ln v_0 = -2\delta t \Rightarrow v = v_0 e^{-2\delta \cdot t}$

再次积分得 $x = \frac{v_0}{2\delta} + x_0 - \frac{v_0}{2\delta} e^{-2\delta t}$

速度，位移呈指数衰减 $e^{-2\delta \cdot t}$

阻尼振动的动力学方程*

物体大致的运动情况:

只有弹性力 $\longrightarrow$ 简谐振动 描述振动: $e^{i\omega t}$ 或 $e^{-i\omega t}$
加粘滞阻力后 $\longrightarrow$ 振幅逐渐变小, 即有指数衰减 $e^{-2\delta \cdot t}$

对动力学方程:
$$\frac{d^2 x}{dt^2} + 2\delta \frac{dx}{dt} + \omega_0^2 x = 0$$

考虑引入试探解 $e^{\gamma t}$ 其中 γ 为复数

代入方程后得到

$$(\gamma^2 + 2\delta\gamma + \omega_0^2)e^{\gamma t} = 0 \longrightarrow \gamma^2 + 2\delta\gamma + \omega_0^2 = 0$$

Q值（品质因子，quality factor）：

由于阻尼的存在，物体在每完成一个周期的振荡，其总能量 E 就衰减去 $(\Delta E)_T$

定义品质因子Q，用来度量阻尼振动中保持能量的能力：

$$Q = 2\pi \frac{E}{(\Delta E)_T} \qquad Q = 2\pi \frac{kA_0^2 / 2}{kA_0^2 / 2 - kA_0^2 e^{-2\delta t} / 2} = \frac{2\pi}{1 - e^{-2\delta t}}$$

Q值越大，则阻尼和能量损耗越小。

Q值可能随时间改变，但大部分的阻尼振动中，Q值近似与时间无关，是个常数。

H. 受迫振动 (forced oscillation)

受迫振动： 系统在周期性的外力持续作用下所发生的振动。

驱动力： 周期性的外力

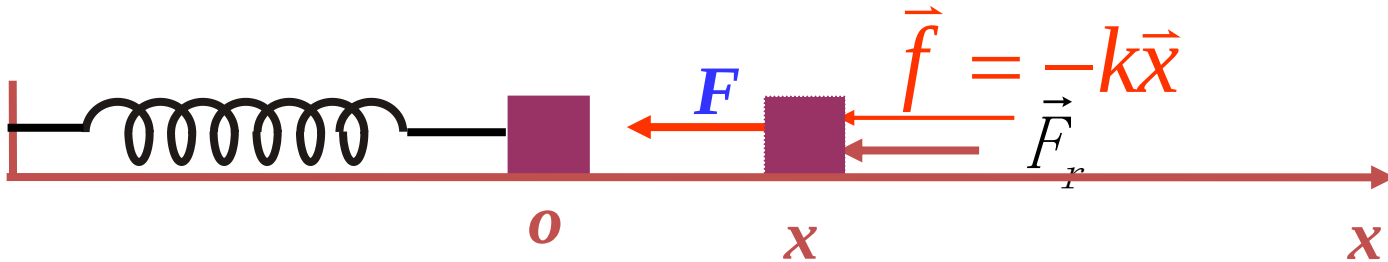
$$F = F_0 \cos \omega_d t$$

运动方程：

$$m \frac{d^2 x}{dt^2} = -kx - \gamma \frac{dx}{dt} + F_0 \cos \omega_d t$$

$$\text{设 } \omega_0^2 = \frac{k}{m}, \quad \delta = \frac{\gamma}{2m}$$

$$\frac{d^2 x}{dt^2} + 2\delta \frac{dx}{dt} + \omega_0^2 x = \frac{F_0}{m} \cos \omega_d t$$

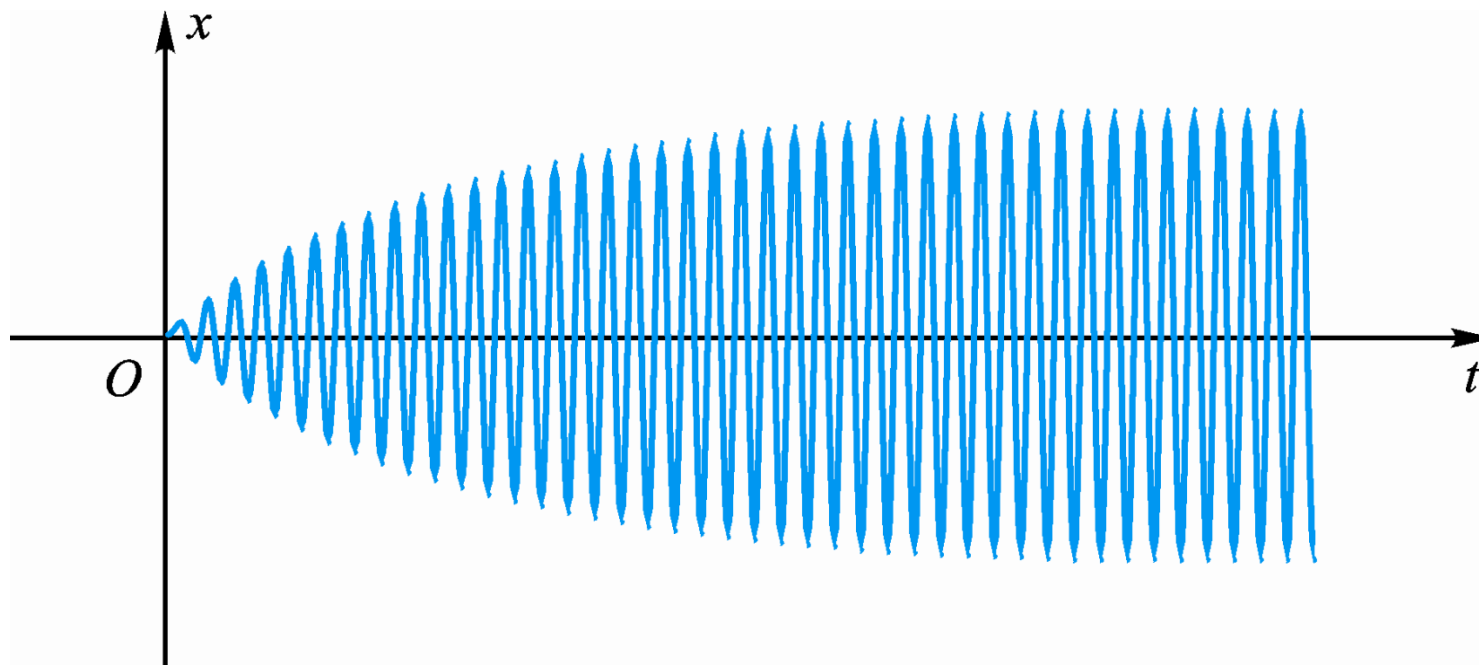


注意：新的周期不一定等于简谐振动系统的本征周期。

当阻尼较小, $\delta < \omega_0$ 时, 方程的解:

$$x = A_0 e^{-\delta t} \cos(\sqrt{\omega_0^2 - \delta^2} t + \varphi'_0) + A \cos(\omega_d t + \varphi)$$

稳定振动状态: $x = A \cos(\omega t + \varphi)$



$$x = A \cos(\omega_d t + \varphi)$$

在稳定振动状态下，受迫振动的频率等于驱动力的频率。

$$A = \frac{F_0}{m \sqrt{(\omega_0^2 - \omega_d^2)^2 + 4\delta^2 \omega_d^2}}$$

$$\tan \varphi = -\frac{2\delta \omega_d}{\omega_0^2 - \omega_d^2}$$

稳态时振动物体速度：

$$v = \frac{dx}{dt} = v_m \cos(\omega_d t + \varphi + \frac{\pi}{2})$$

$$v_m = \frac{\omega_d F_0}{m \sqrt{(\omega_0^2 - \omega_d^2)^2 + 4\delta^2 \omega_d^2}}$$

在受迫振动中，周期性的驱动力对振动系统提供能量，另一方面系统又因阻尼而消耗能量，若二者相等，则系统达到稳定振动状态。

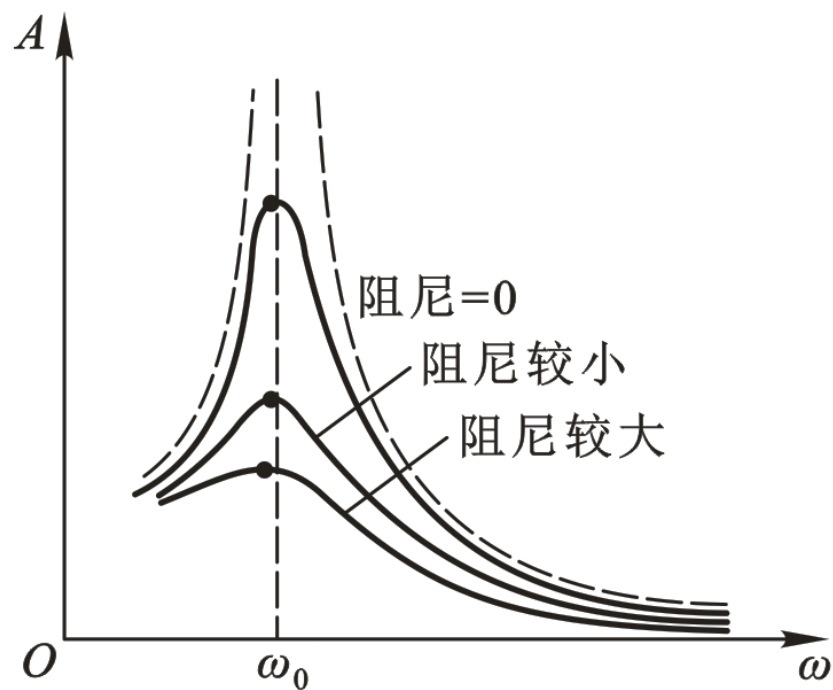
2、共振

$$A = \frac{F_0}{m\sqrt{(\omega_0^2 - \omega_d^2)^2 + 4\delta^2\omega_d^2}}$$

当驱动力的角频率等于某个特定值时，位移振幅达到最大值的现象称为**位移共振** (displacement resonance) 。

$$\frac{dA}{d\omega_d} = 0$$

$$\Rightarrow \omega_{\text{共振}} = \sqrt{\omega_0^2 - 2\delta^2}$$



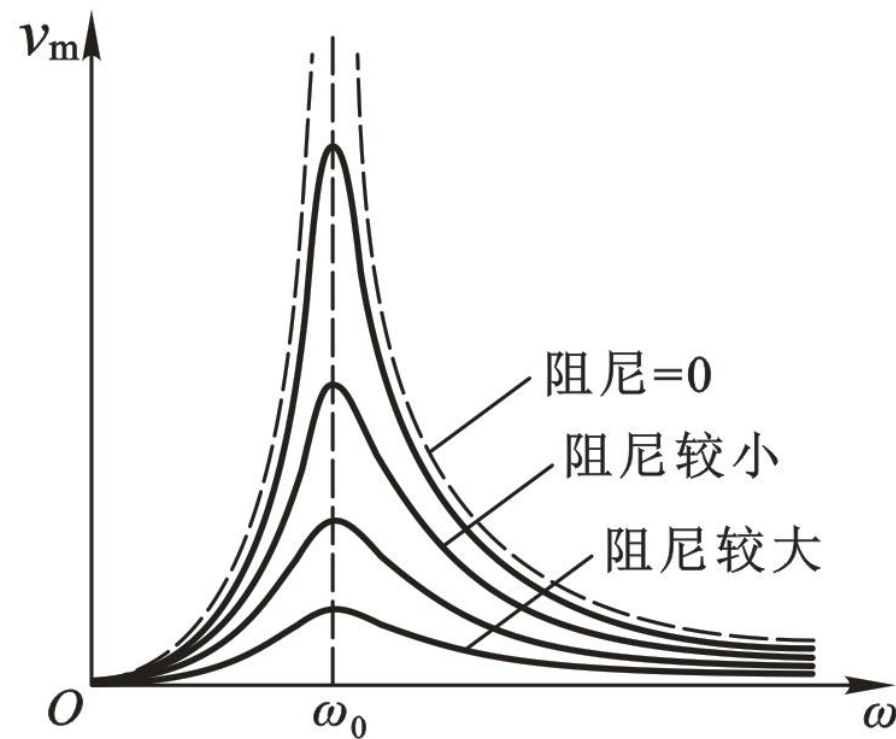
$$v_m = \frac{\omega_d F_0}{m \sqrt{(\omega_0^2 - \omega_d^2)^2 + 4\delta^2 \omega_d^2}}$$

受迫振动速度在一定条件下发生共振的现象称为速度共振（velocity resonance）。

$$\frac{dv_m}{d\omega_d} = 0$$

$$\Rightarrow \omega_{\text{共振}} = \omega_0$$

在阻尼很小的前提下，速度共振和位移共振可以认为等同。



I. 简谐振动的合成

(a) : 同一直线上两个同频率谐振动的合成

某一质点同时参与两个独立的、同方向、同频率的简谐振动，其振动位移分别为

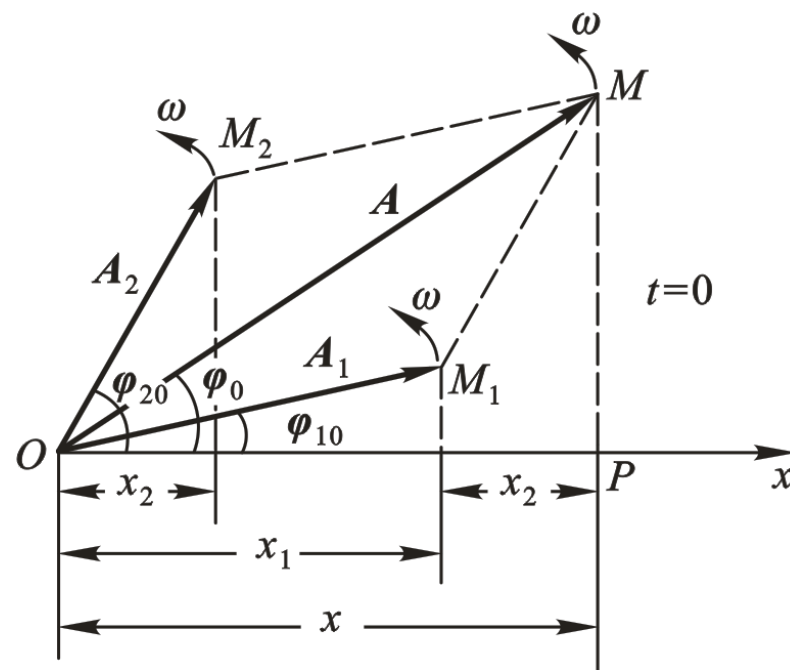
$$x_1 = A_1 \cos(\omega t + \varphi_{10}) \quad x_2 = A_2 \cos(\omega t + \varphi_{20})$$

合振动：（由振动的叠加原理）

$$x = x_1 + x_2$$

$$\vec{A} = \vec{A}_1 + \vec{A}_2$$

$$x = A \cos(\omega t + \varphi_0)$$

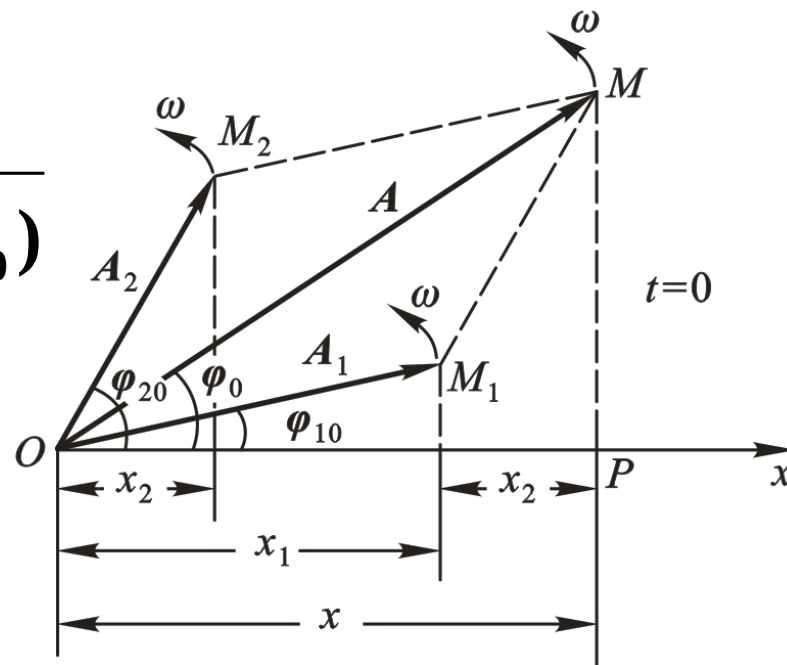


➤ 合振动仍为同方向同频率的简谐振动。

合振动: $x = A \cos(\omega t + \varphi_0)$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_{20} - \varphi_{10})}$$

$$\varphi = \arctan \frac{A_1 \sin \varphi_{10} + A_2 \sin \varphi_{20}}{A_1 \cos \varphi_{10} + A_2 \cos \varphi_{20}}$$



(1) 若 $\varphi_{20} - \varphi_{10} = 2k\pi$ ($k = 0, \pm 1, \pm 2, \dots$)

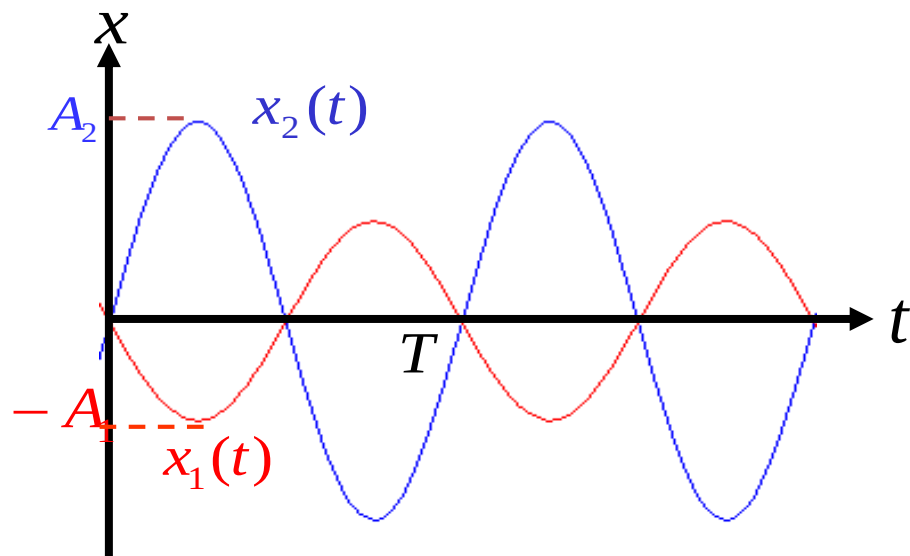
则 $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2} = A_1 + A_2$

(2) 若 $\varphi_{20} - \varphi_{10} = (2k + 1)\pi$ ($k = 0, \pm 1, \pm 2, \dots$)

则 $A = \sqrt{A_1^2 + A_2^2 - 2A_1A_2} = |A_1 - A_2|$

[例题4-8]两个同方向的简谐振动曲线如图所示，求：

1. 合振动的振幅；
2. 合振动的振动式。



(b) : 同一直线上两个不同频率谐振动的合成 拍

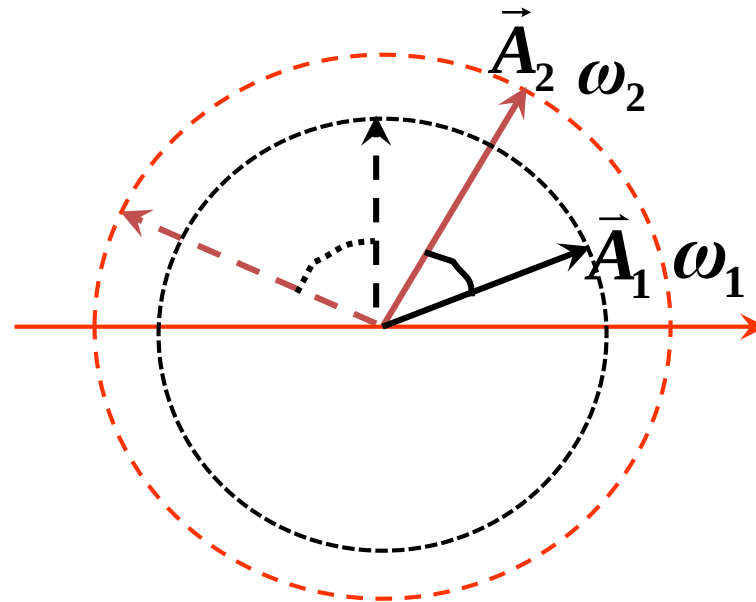
设同方向、角频率分别为 ω_1 和 ω_2 的两简谐振动 ($\omega_2 > \omega_1$) , 它们所对应的旋转矢量分别为 $\vec{A}_1$ 和 $\vec{A}_2$

$$x_1 = A_1 \cos(\omega_1 t + \varphi_1)$$

$$x_2 = A_2 \cos(\omega_2 t + \varphi_2)$$

$$x = x_1 + x_2$$

$$= A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$$



设： $A_1 = A_2 = A$ $\varphi_1 = \varphi_2 = \varphi_0$

$$x = 2A \cos \frac{\omega_2 - \omega_1}{2} t \cdot \cos \left(\frac{\omega_2 + \omega_1}{2} t + \varphi_0 \right)$$

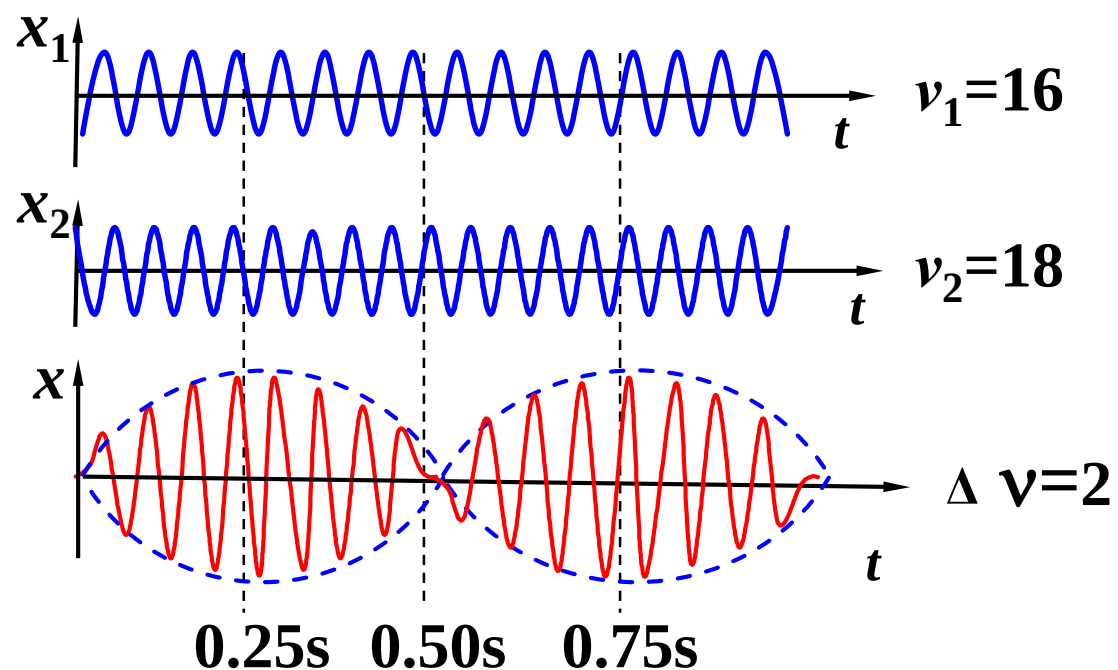
振幅： $\left| 2A \cos \frac{\omega_2 - \omega_1}{2} t \right|$

谐振因子： $\cos \left(\frac{\omega_2 + \omega_1}{2} t + \varphi_0 \right)$

当 ω_2 、 ω_1 满足关系： $|\omega_2 - \omega_1| \ll \omega_1, \omega_2$

振幅随时间缓慢变化，谐振因子角频率 $\frac{\omega_2 + \omega_1}{2} \approx \omega_1$

合振幅时强时弱周期性缓慢变化的现象 拍现象



拍——频率较大但相差不大的两个同方向简谐振动合成时产生合振动振幅周期性变化的现象.

拍频——单位时间内振动加强或减弱的次数.

拍的周期:

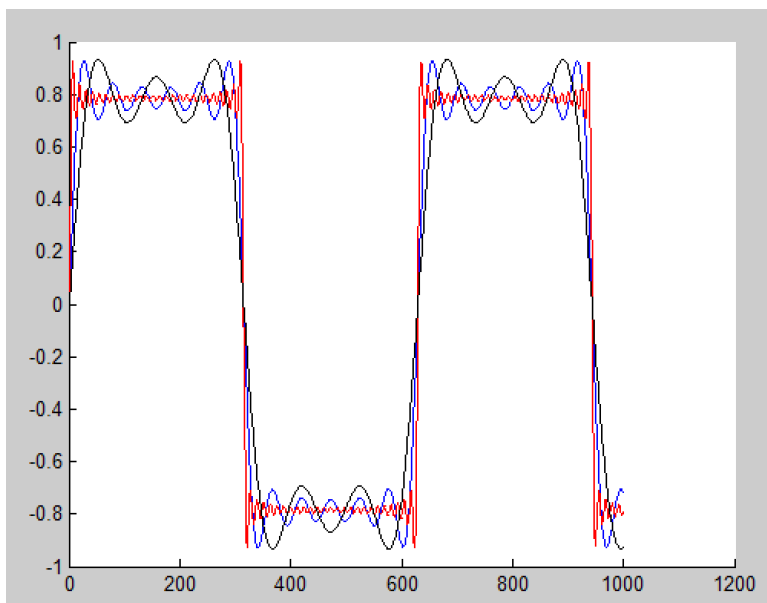
$$\tau = \frac{2\pi}{|\omega_2 - \omega_1|}$$

拍的频率:

$$\nu_b = \frac{|\omega_2 - \omega_1|}{2\pi} = |\nu_2 - \nu_1|$$

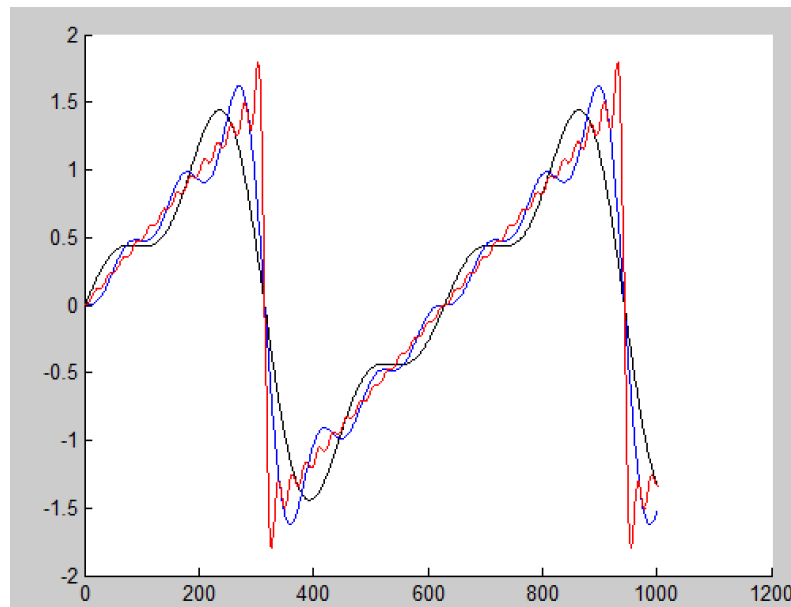
非简谐振动

任何一个周期性运动，都能用其频率整数倍的简谐振动的叠加得到。



$$(\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \frac{1}{7} \sin 7\omega t + \cdots)$$

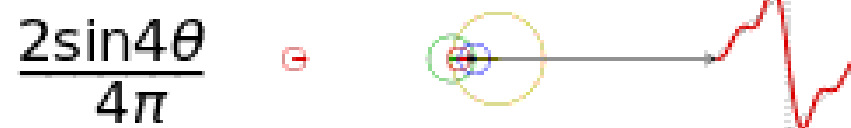
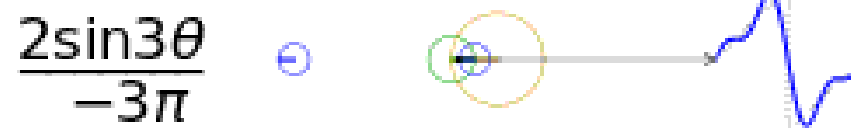
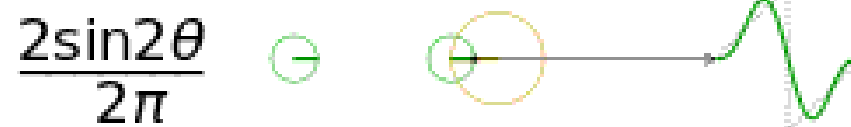
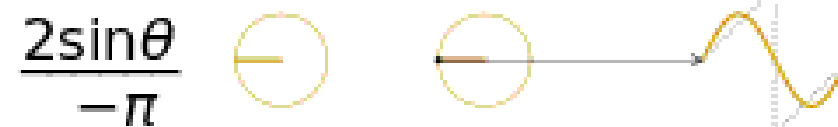
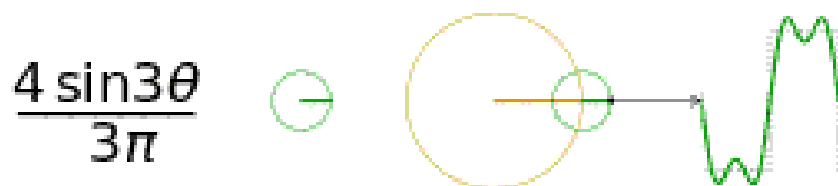
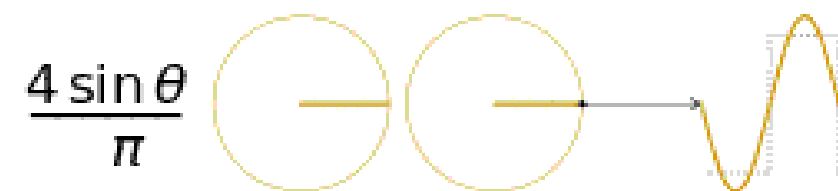
三项，六项，二十六项的叠加结果



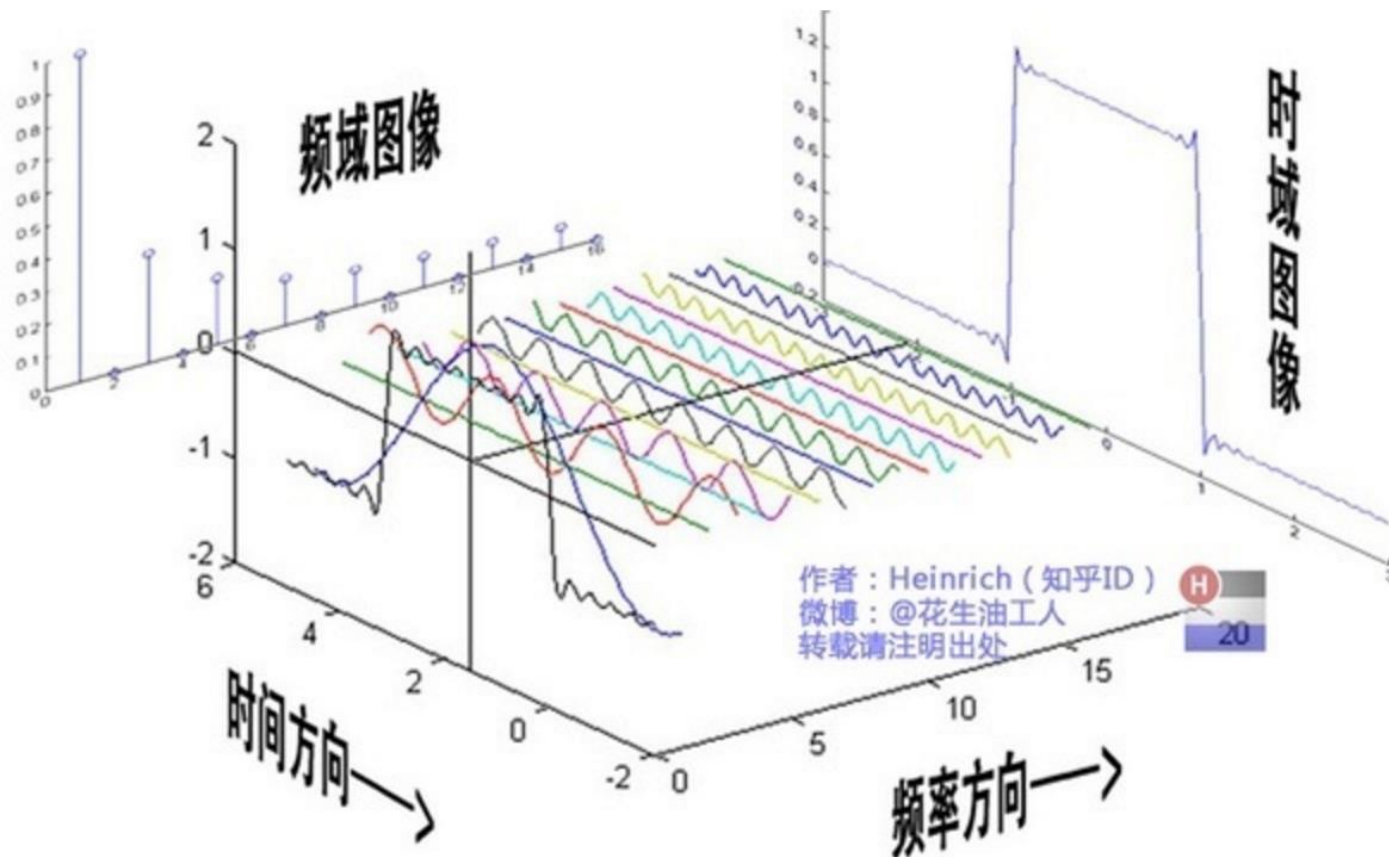
$$\sin(\omega t) - \frac{1}{2} \sin(2\omega t) + \frac{1}{3} \sin(3\omega t) - \cdots$$

三项，六项，二十六项的叠加结果

非简谐振动-y方向投影示意



傅里叶级数



一个信号，如果用一组简谐振动将其表示出来，这组简谐振动称为**傅里叶级数**。

将信号分解有什么用？

背后的数学问题：

1. 是否对所有的信号都能做这样的函数展开（**傅里叶级数的完备性**）。
2. 展开的级数之和，是否能够收敛到原来的函数（**级数的收敛性**）。

(c) : 垂直方向, 李萨如图形

1. 频率相同

两谐振动

$$x = A_1 \cos(\omega_0 t + \alpha_1)$$

$$y = A_2 \cos(\omega_0 t + \alpha_2)$$

消去 t

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy}{A_1 A_2} \cos(\alpha_2 - \alpha_1) = \sin^2(\alpha_2 - \alpha_1)$$

合成轨迹一般为一椭圆, 两振幅相等时为圆. 具体的来说: 形状由相差决定.

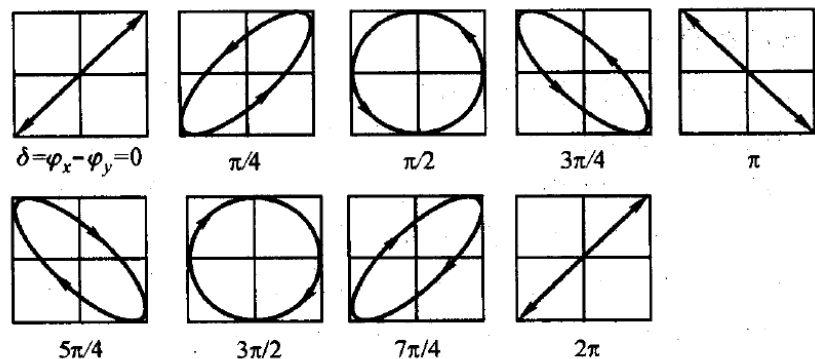
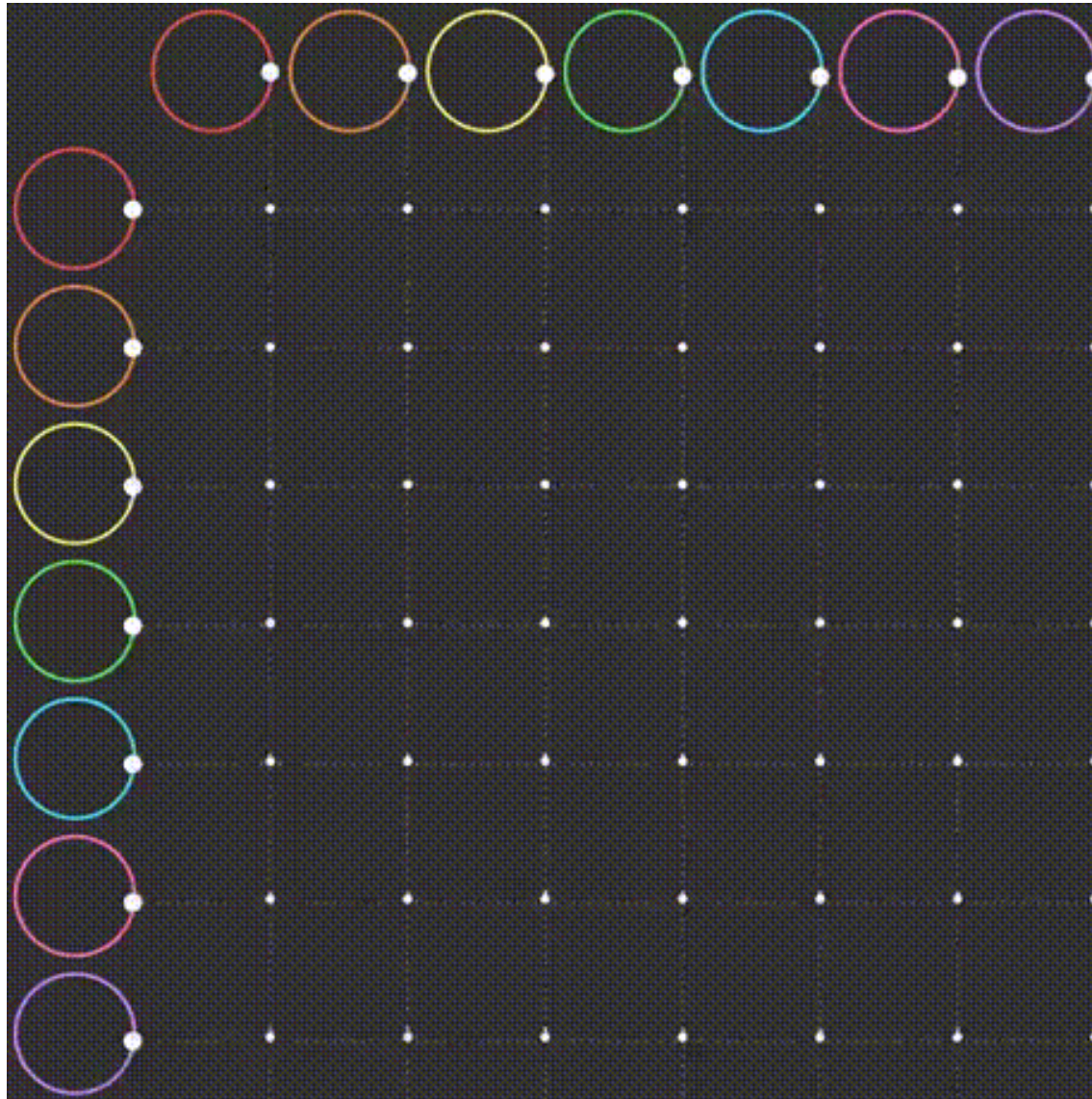


图 8.2-6 不同相位差对应的椭圆

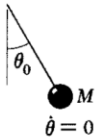
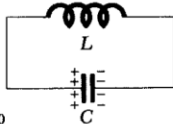
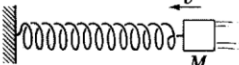
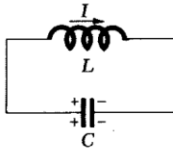
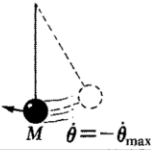
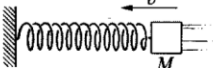
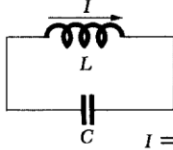
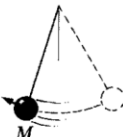
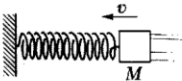
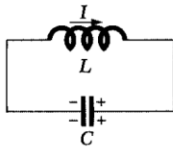
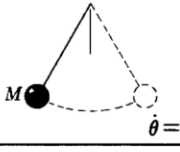
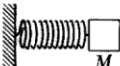
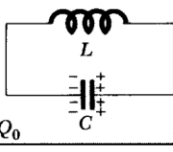
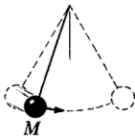
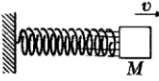
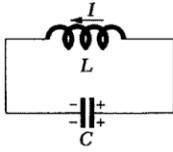
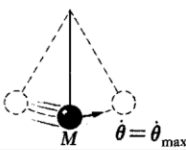
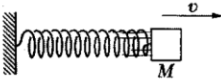
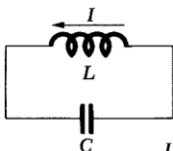
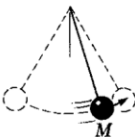
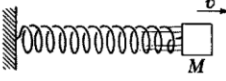
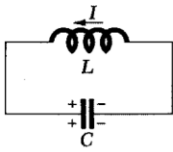
说明: 任何一个直线简谐振动, 椭圆运动或匀速圆周运动都可分解为两个相互垂直的简谐振动.

2. 频率不同，为整数比——李萨如图形



伯克利物理学教程

单摆、弹簧、LC振荡 随时间的变化

Simple pendulum	Mass-spring system	LC circuit	Kinetic energy, K	Potential energy, U
A $t = 0$  $\theta = \theta_0$ $\dot{\theta} = 0$	$v = 0$  $x = x_0$	 $Q = Q_0$ $I = 0$		
B $t = \frac{\pi}{4\omega}$  M	 M	 C		
C $t = \frac{\pi}{2\omega}$  M $\dot{\theta} = -\dot{\theta}_{\max}$	$v = -v_{\max}$  M $x = 0$	 C $I = -I_{\max}$		
D $t = \frac{3\pi}{4\omega}$  M	 M	 C		
E $t = \frac{\pi}{\omega}$  M $\dot{\theta} = 0$	$v = 0$  M $x = -x_0$	 C $I = 0$		
F $t = \frac{5\pi}{4\omega}$  M	 M	 C		
G $t = \frac{3\pi}{2\omega}$  M $\dot{\theta} = \dot{\theta}_{\max}$	$v = v_{\max}$  M $x = 0$	 C $I = I_{\max}$		
H $t = \frac{7\pi}{4\omega}$  M	 M	 C		

阻尼振动的动力学方程的解 1*

我们讨论**特征方程** $\gamma^2 + 2\delta\gamma + \omega_0^2 = 0$ 解的情况

δ 称为阻尼系数

$\gamma^2 + 2\delta\gamma + \delta^2 = \delta^2 - \omega_0^2$ 讨论判别式 $\Delta = \delta^2 - \omega_0^2$ 的正负性

1) $\Delta > 0$, 即 $\delta > \omega_0$ (**过阻尼**)

特征方程的两个实数根分别为 $r_1 = -\delta + \sqrt{\delta^2 - \omega_0^2}$, $r_2 = -\delta - \sqrt{\delta^2 - \omega_0^2}$

方程通解为

$$Ae^{r_1 t} + Be^{r_2 t} = e^{-\delta t} (Ae^{t\sqrt{\delta^2 - \omega_0^2}} + Be^{-t\sqrt{\delta^2 - \omega_0^2}})$$

阻尼较大时 $\delta > \omega_0$, 质点缓慢回到平衡位置,
不作往复运动。

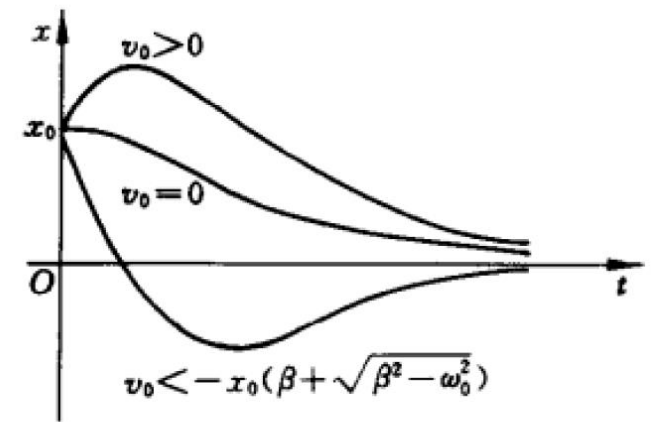
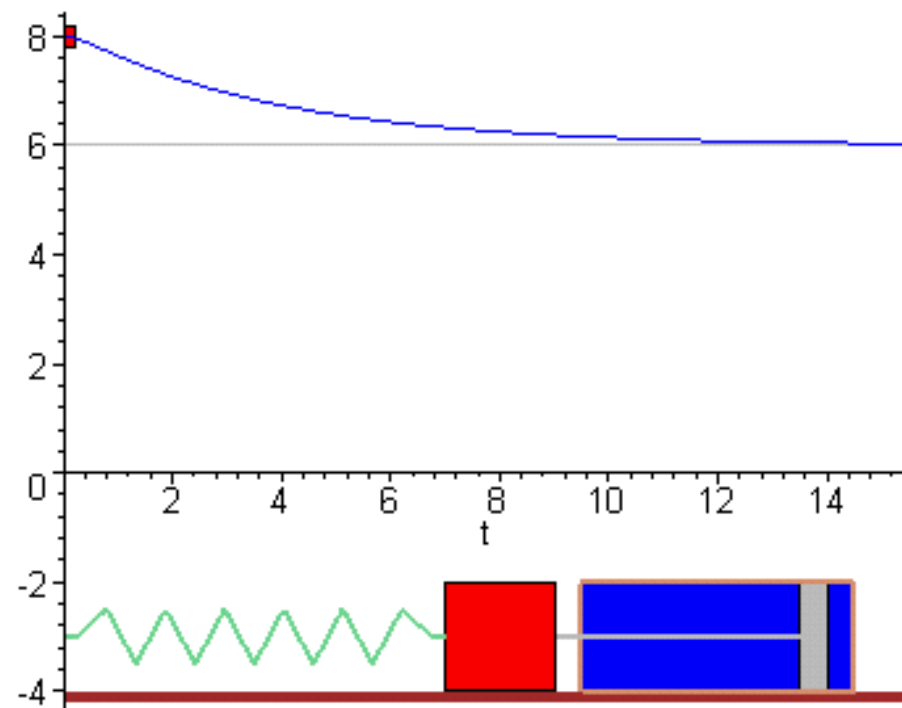
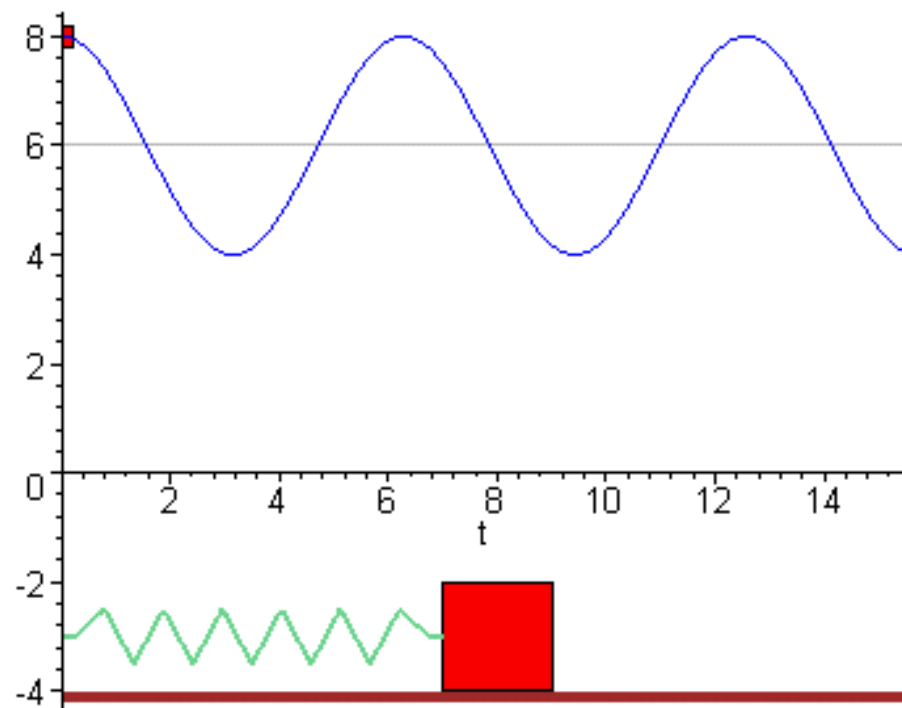


图 7-43 过阻尼

过阻尼振动*



阻尼振动的动力学方程的解 2*

2) $\Delta=0$, 即 $\delta=\omega_0$ (**临界阻尼**)

特征方程只有一个实数根分别为 $r_1 = r_2 = -\delta$

因此找到一个解为 $e^{-\delta t}$, **猜测**另一个独立的解为 $te^{-\delta t}$, 代入验算后成立。

方程通解为 $Ae^{-\delta t} + Bte^{-\delta t} = e^{-\delta t}(A + Bt)$

当 ($\delta = \omega_0$) 时, 为“临界阻尼”情况, 是质点不作往复运动的一个极限

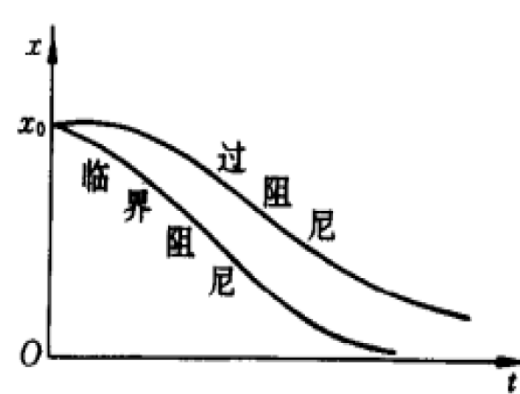
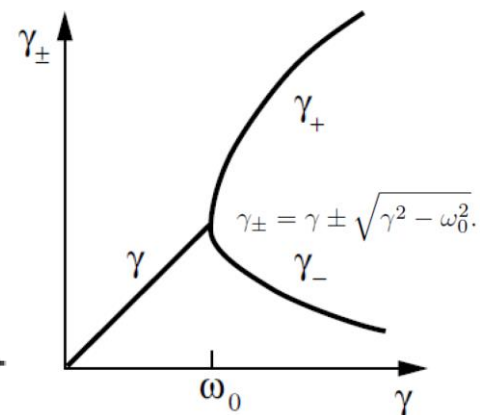
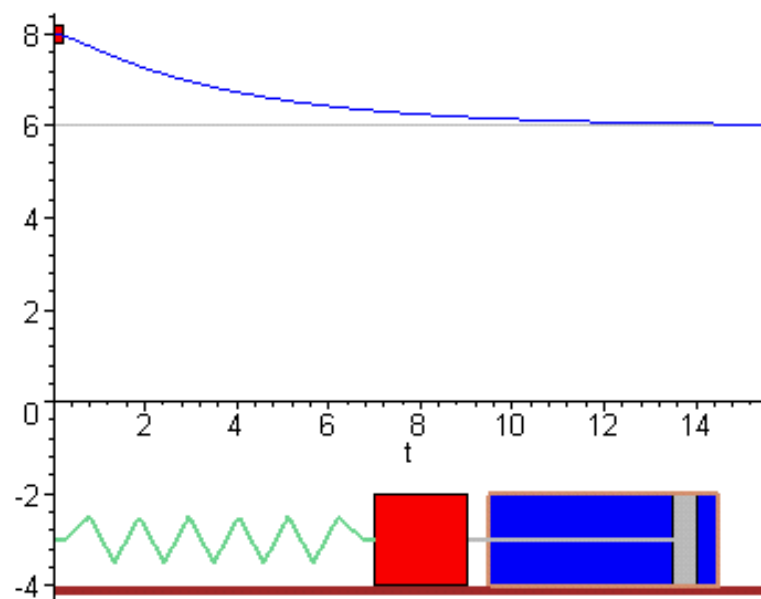


图 7-44 临界阻尼

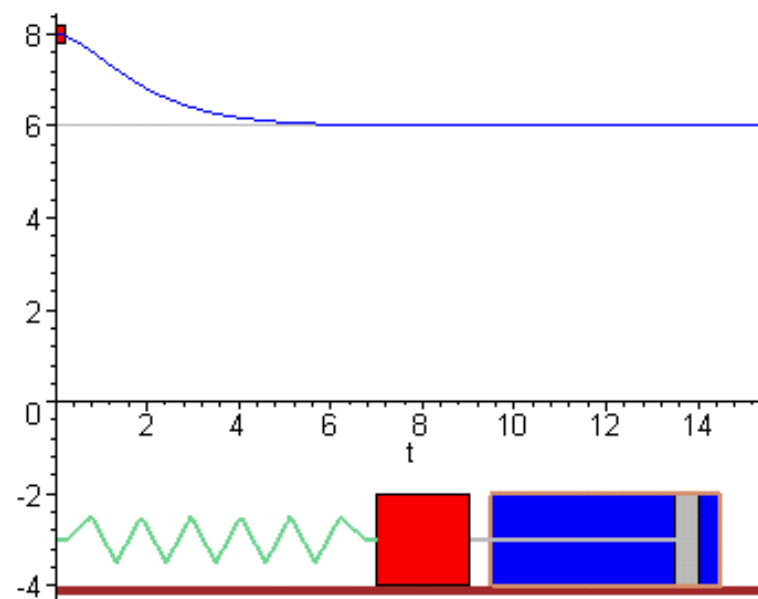


临界阻尼时振幅衰减最快

临界阻尼振动*



过阻尼



临界阻尼

阻尼振动的动力学方程的解*

3) $\Delta > 0$, 即 $\delta < \omega_0$ (**低阻尼**)

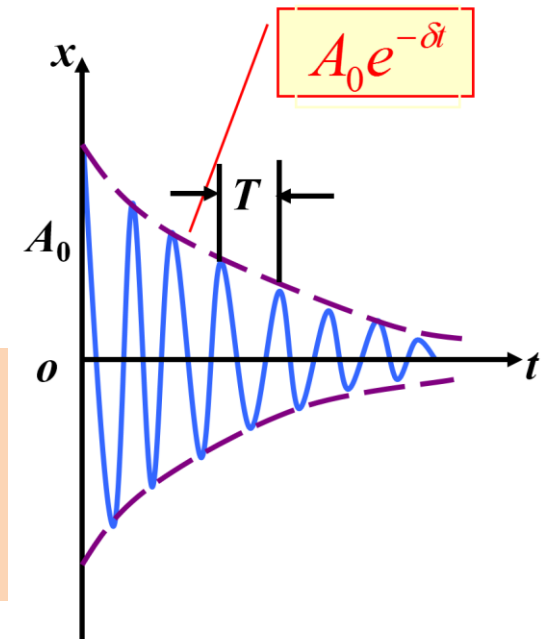
特征方程的两个复数根分别为 $r_1 = -\delta + i\sqrt{\omega_0^2 - \delta^2}$, $r_2 = -\delta - i\sqrt{\omega_0^2 - \delta^2}$

方程通解为 $Ae^{r_1 t} + Be^{r_2 t} = e^{-\delta t}(Ae^{it\sqrt{\omega_0^2 - \delta^2}} + Be^{-it\sqrt{\omega_0^2 - \delta^2}})$ (A,B为复数, 有物理意义的是解的实数部分)

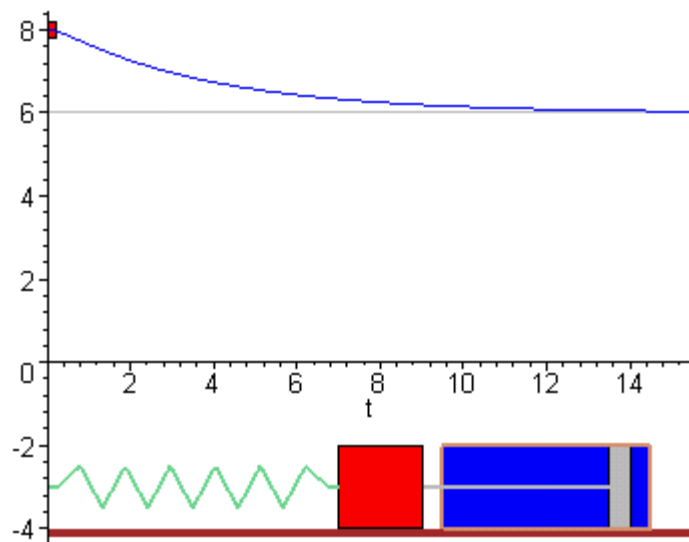
将其组合出我们熟悉的 $\cos\omega t$, $\sin\omega t$ 的形式, 我们得到方程的另一个形式通解为

$$A'e^{r_1 t} + B'e^{r_2 t} = e^{-\delta t}(A' \cos(t\sqrt{\omega_0^2 - \delta^2}) + B' \sin(t\sqrt{\omega_0^2 - \delta^2}))$$

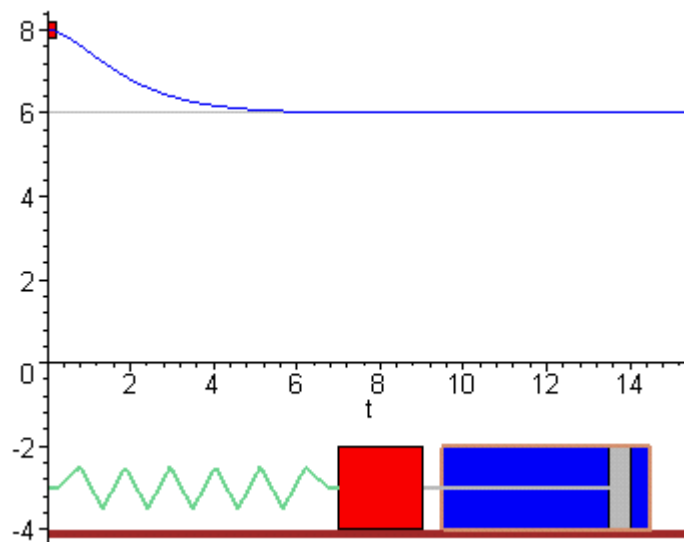
阻尼较小时 ($\delta < \omega_0$), 振动为减幅振动, 振幅 $Ae^{-\delta t}$ 随时间按指数规律迅速减少。阻尼越大, 减幅越迅速。振动周期大于自由振动周期。



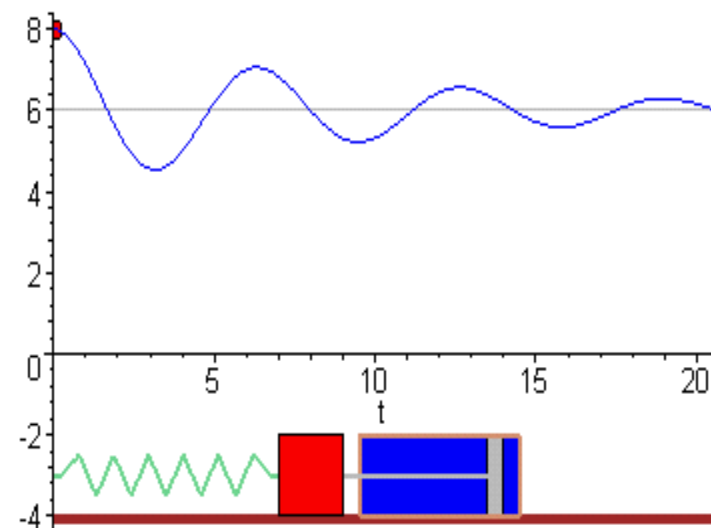
阻尼振动*



过阻尼



临界阻尼



欠阻尼